

Preparing Activity: NAVFAC

Superseding
UFGS-23 52 00 (April 2008)
UFGS-23 52 53.00 20 (April 2006)

UNIFIED FACILITIES GUIDE SPECIFICATIONS

References are in agreement with UMRL dated July 2026

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DIVISION 23 - HEATING, VENTILATING, AND AIR CONDITIONING (HVAC)

SECTION 23 52 00.04

STEAM BOILERS AND EQUIPMENT (OVER 6,000,000 BTU/HR) INPUT

08/26

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SECTION 23 52 00.04

STEAM BOILERS AND EQUIPMENT (OVER 6,000,000 BTU/HR) INPUT 08/26

NOTE: This guide specification covers the requirements for packaged steam boilers and equipment with gross inputs over **5275 kw 6,000,000 Btu's** per hour and is intended for new construction boiler plant installations.

Adhere to UFS 1-300-02 Unified Facilities Guide Specifications (UFGS) Format Standard when editing this guide specification or preparing new project specification sections. Edit this guide specification for project specific requirements by adding, deleting, or revising text. For bracketed items, choose applicable item(s) or insert appropriate information.

Remove information and requirements not required in respective project, whether or not brackets are present.

Comments, suggestions and recommended changes for this guide specification are welcome and should be submitted as a Criteria Change Request (CCR).

NOTE: These boilers are equipped for oil-, gas-, or oil-and-gas combination firing, as specified.

NOTE: The following information must be shown on the project drawings:

1. Dimensions of construction
2. Relationship of materials

3. Quantities, location and capacity of equipment.

PART 1 GENERAL

1.1 REFERENCES

NOTE: This paragraph is used to list the publications cited in the text of the guide specification. The publications are referred to in the text by basic designation only and listed in this paragraph by organization, designation, date, and title.

Use the Reference Wizard's Check Reference feature when you add a Reference Identifier (RID) outside of the Section's Reference Article to automatically place the reference in the Reference Article. Also use the Reference Wizard's Check Reference feature to update the issue dates.

References not used in the text will automatically be deleted from this section of the project specification when you choose to reconcile references in the publish print process.

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ACOUSTICAL SOCIETY OF AMERICA (ASA)

ASA S1.13 (2020; R 2025) American National Standard
Measurement of Sound Pressure Levels in Air

AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)

AMCA 801 (2001; R 2008) Industrial Process/Power
Generation Fans: Specification Guidelines

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 360 (2022) Specification for Structural Steel
Buildings

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z21.13/CSA 4.9 (2022) Gas-Fired Low Pressure Steam and
Hot Water Boilers

AMERICAN PETROLEUM INSTITUTE (API)

API STD 610 (2010; Errata 2011) Centrifugal Pumps for
Petroleum, Petrochemical, and Natural Gas
Industries

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING
ENGINEERS (ASHRAE)

ASHRAE 52.2 (2025) Method of Testing General
Ventilation Air-Cleaning Devices for
Removal Efficiency by Particle Size

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.20 (2023) Metallic Gaskets for Pipe Flanges

ASME B40.100 (2022) Pressure Gauges and Gauge
Attachments

ASME BPVC (2025) Boiler and Pressure Vessels Code

ASME BPVC SEC I (2017) BPVC Section I-Rules for
Construction of Power Boilers

ASME BPVC SEC IV (2017) BPVC Section IV-Rules for
Construction of Heating Boilers

ASME BPVC SEC VIII D1 (2023) BPVC Section VIII-Rules for
Construction of Pressure Vessels Division 1

ASME CSD-1 (2024) Control and Safety Devices for
Automatically Fired Boilers

ASME PTC 4 (2013; R 2023) Fired Steam Generators

ASME PTC 10 (2022) Performance Test Code on
Compressors and Exhausters

ASME PTC 19.5 (2022) Flow Measurement

ASME PTC 19.11 (2008; R 2013) Steam and Water Sampling,
Conditioning, and Analysis in the Power
Cycle

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M (2025) Structural Welding Code - Steel

AWS Z49.1 (2021) Safety in Welding, Cutting and
Allied Processes

ASTM INTERNATIONAL (ASTM)

ASTM A283/A283M (2024) Standard Specification for Low and
Intermediate Tensile Strength Carbon Steel
Plates

ASTM A653/A653M (2025a) Standard Specification for Steel
Sheet, Zinc-Coated (Galvanized) or
Zinc-Iron Alloy-Coated (Galvannealed) by
the Hot-Dip Process

ASTM B88 (2022) Standard Specification for Seamless
Copper Water Tube

ASTM B88M	(2020) Standard Specification for Seamless Copper Water Tube (Metric)
ASTM B117	(2025) Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM C27	(1998; R 2022) Fireclay and High-Alumina Refractory Brick
ASTM C34	(2023) Standard Specification for Structural Clay Loadbearing Wall Tile
ASTM C155	(1997; R 2022) Standard Specification for Insulating Firebrick
ASTM C401	(2012; R 2022) Alumina and Alumina-Silicate Castable Refractories
ASTM D396	(2021) Standard Specification for Fuel Oils
ASTM D596	(2001; R 2018) Standard Guide for Reporting Results of Analysis of Water
ASTM D888	(2012; E 2013) Dissolved Oxygen in Water
ASTM D1066	(2018; E 2018) Standard Practice for Sampling Steam
ASTM D2186	(2005; R 2009) Deposit-Forming Impurities in Steam
ASTM F1097	(2017; R 2022) Standard Specification for Mortar, Refractory (High-Temperature, Air-Setting)
ASTM F1139	(1988; R 2024) Steam Traps and Drains

COMPRESSED AIR AND GAS INSTITUTE (CAGI)

CAGI B19.1	(2010) Safety Standard for Compressor Systems
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FM GLOBAL (FM)

FM APP GUIDE	(updated on-line) Approval Guide https://www.approvalguide.com/
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MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

MSS SP-58	(2025) Pipe Hangers and Supports - Materials, Design and Manufacture, Selection, Application, and Installation
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NATIONAL BOARD OF BOILER AND PRESSURE VESSEL INSPECTORS (NBBI)

NBBI NB-27	(1991) National Board Rules and Recommendations for the Design and
------------	--

Construction of Boiler Blowoff Systems

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA ICS 6 (1993; R 2016) Industrial Control and Systems: Enclosures

NEMA MG 1 (2021) Motors and Generators

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 31 (2024; TIA 23-1) Standard for the Installation of Oil-Burning Equipment

NFPA 54 (2024) National Fuel Gas Code

NFPA 70 (2026; TIA 26-1; ERTA 26-1; TIA 26-2; TIA 26-3; TIA 26-4; TIA 26-5; TIA 26-6; TIA 26-7; ERTA 26-2; ERTA 26-3) National Electrical Code

NFPA 85 (2025) Boiler and Combustion Systems Hazards Code

NFPA 211 (2019) Standard for Chimneys, Fireplaces, Vents, and Solid Fuel-Burning Appliances

SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)

SAE AMS3126D (2021) Aluminum Coating Material, Corrosion and Heat Resistant Thermosetting, Inorganic Binder

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-430-07 2003; with Change 5, 2019) Operations and Maintenance: Inspection and Certification of Boilers and Unfired Pressure Vessels - INACTIVE

U.S. DEPARTMENT OF ENERGY (DOE)

Energy Star Energy Star Energy Efficiency Labeling System (FEMP)

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910-SUBPART D Walking - Working Surfaces

UL SOLUTIONS (UL)

UL 296 (2017; Reprint Jan 2026) UL Standard for Safety Oil Burners

UL 726 (1995; Reprint Sep 2024) UL Standard for Safety Oil-Fired Boiler Assemblies

UL 795 (2016; Reprint May 2022) UL Standard for Safety Commercial-Industrial Gas Heating

Equipment

UL FLAMMABLE & COMBUSTIBLE

(2012) Flammable and Combustible Liquids and Gases Equipment Directory

1.2 SYSTEM DESCRIPTION

Describe the performance or design requirements and tolerances of the complete system. Section 23 52 33.01 STEAM HEATING PLANT WATERTUBE COAL/OIL OR COAL.

1.2.1 Heating Surface and Volume Measurements

Submit heating surface and volume measurements, including heat release calculations sufficient to establish compliance of boilers with heat release requirements. Base calculations on the specified efficiency and capacity.

1.3 RELATED REQUIREMENTS

The following guide specification sections apply to this section with the additions and modifications as stated in the paragraph cited:

NOTE: UFGS Sections that have been unified for use by all participating agencies have a level 3 or level 4 MasterFormat™ number. UFGS Sections that are agency-specific have a fifth level number indicating it as an agency-specific specification section. A Section that has a fifth level number "10" indicates USACE, a "20" indicates NAVFAC, and a "30" indicates AFCEC. The Preparing Activity is indicated in the banner of each UFGS Section.

Remove Sections not required for project.

01 78 23 OPERATION AND MAINTENANCE DATA

[01 78 24.00 20 FACILITY DATA WORKBOOK (FDW)

] 03 30 00 CAST-IN-PLACE CONCRETE

05 12 00 STRUCTURAL STEEL

09 90 00 PAINTS AND COATINGS

09 97 13.15 20 LOW VOC FLEXIBLE INTERIOR COATING OF STEEL FUEL TANKS

09 97 13.28 PROTECTION OF BURIED STEEL PIPING AND BULKHEAD TIE RODS

11 82 19 PACKAGED INCINERATORS

13 48 73 SEISMIC CONTROL FOR NONSTRUCTURAL COMPONENTS

21 13 13 WET PIPE SPRINKLER SYSTEMS, FIRE PROTECTION

22 00 00 PLUMBING, GENERAL PURPOSE

- 23 05 15 COMMON PIPING FOR HVAC
- 23 05 48.19 SEISMIC BRACING FOR MECHANICAL SYSTEMS
- 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS
- 23 09 13 INSTRUMENTATION AND CONTROL DEVICES FOR HVAC
- [23 21 13.00 20 LOW TEMPERATURE WATER (LTW) HEATING SYSTEM
-][23 21 13.23 20 [HIGH][MEDIUM] TEMPERATURE WATER SYSTEMS WITHIN BUILDINGS
-][23 22 26.00 20 STEAM SYSTEM AND TERMINAL UNITS
-] 23 25 00 CHEMICAL TREATMENT OF WATER FOR MECHANICAL SYSTEMS
- [23 57 10.00 10 FORCED HOT WATER HEATING SYSTEMS USING WATER AND STEAM HEAT EXCHANGERS
-] 26 20 00 INTERIOR DISTRIBUTION SYSTEM
- [40 17 26.00 20 WELDED PRESSURE PIPING
-] 41 22 13.14 BRIDGE CRANES, OVERHEAD ELECTRIC, TOP RUNNING
- 44 10 00 AIR POLLUTION CONTROL

1.4 ELECTRICAL REQUIREMENTS

Furnish motors, controllers, disconnects and contactors with their respective pieces of equipment. Motors, controllers, disconnects and contactors must conform to and have electrical connections provided under Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Furnish internal wiring for components of packaged equipment as an integral part of the equipment. Extended voltage range motors will not be permitted. Controllers and contactors must have a maximum of 120 volt control circuits, and must have auxiliary contacts for use with the controls furnished. When motors and equipment furnished are larger than sizes indicated, the cost of additional electrical service and related work must be included under the section that specified that motor or equipment. Power wiring and conduit for field installed equipment must be provided under and conform to the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

1.5 SUBMITTALS

NOTE: Review Submittal Description (SD) definitions in Section 01 33 00 SUBMITTAL PROCEDURES and edit the following list, and corresponding submittal items in the text, to reflect only the submittals required for the project. The Guide Specification technical editors have classified those items that require Government approval, due to their complexity or criticality, with a "G." Generally, other submittal items can be reviewed by the Contractor's Quality Control System. Only add a "G" to an item if the submittal is sufficiently important or complex in context of the project.

For Army projects, fill in the empty brackets following the "G" classification with a code of up to three characters to indicate the approving authority. Codes for Army projects using the Resident Management System (RMS) are: "AE" for Architect-Engineer; "DO" for District Office (Engineering Division or other organization in the District Office); "AO" for Area Office; "RO" for Resident Office; and "PO" for Project Office. Codes following the "G" typically are not used for Navy and Air Force projects.

The "S" classification indicates submittals required as proof of compliance for sustainability Guiding Principles Validation or Third Party Certification and as described in Section 01 33 00 SUBMITTAL PROCEDURES.

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Detail Drawings

Steam Boiler System

SD-03 Product Data

Boiler Efficiency

Materials and Equipment

Spare Parts

Heating System Tests

Fuel System Tests

Qualifications

Field Instructions

Tests

SD-05 Design Data

Heating Surface and Volume Measurements

Heat Release Calculations

SD-06 Test Reports

Heating System Tests

Fuel System Tests

Water Treatment Testing

Boiler System Start-Up Tests

SD-07 Certificates

Continuous Emissions Monitoring

Report of Prior Installations

Engineer's Qualification Resume

Start-Up Plan

Start-Up Certification

Boilers

SD-10 Operation and Maintenance Data

Operation and Maintenance Instructions; G, []

Boilers, Data Package 4

SD-11 Closeout Submittals

Indoor Air Quality During Construction; S

1.6 QUALITY ASSURANCE

1.6.1 Material and Equipment Qualifications

Provide materials and equipment that are standard products of manufacturers regularly engaged in the manufacture of such products, which are of a similar material, design and workmanship. Standard products must have been in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year use must include applications of equipment and materials under similar circumstances and of similar size. The product must have been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2 year period.

1.6.2 Alternative Qualifications

Products having less than a two-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturer's factory or laboratory tests, can be shown.

1.6.3 Service Support

The equipment items must be supported by service organizations. Submit a certified list of qualified permanent service organizations for support of the equipment which includes their addresses and qualifications. These service organizations must be reasonably convenient to the equipment installation and able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

1.6.4 Manufacturer's Nameplate

For each item of equipment, provide a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

1.6.5 Definitions

For the International Code Council (ICC) Codes referenced in the contract documents, advisory provisions must be considered mandatory, the word "should" is interpreted as "must." Reference to the "code official" must be interpreted to mean the "Contracting Officer." For Navy owned property, references to the "owner" must be interpreted to mean the "Contracting Officer." For leased facilities, references to the "owner" must be interpreted to mean the "lessor." References to the "permit holder" must be interpreted to mean the "Contractor."

1.6.6 Administrative Interpretations

For ICC Codes referenced in the contract documents, the provisions of Chapter 1, "Administrator," do not apply. These administrative requirements are covered by the applicable Federal Acquisition Regulations (FAR) included in this contract and by the authority granted to the Officer in Charge of Construction to administer the construction of this project. References in the ICC Codes to sections of Chapter 1, must be applied appropriately by the Contracting Officer as authorized by his administrative cognizance and the FAR.

1.6.7 [Report of Prior Installations](#)

Boilers, burners, combustion controls, and feedwater equipment must be of proven design. Submit evidence to show that substantially identical equipment of comparable capacity within 20 percent has been successfully installed and operated in not less than three installations under comparable operating conditions for a period of not less than 2 years.

1.6.8 [Engineer's Qualification Resume](#)

Submit a printed certified qualification resume of the engineer or technician to the Contracting Officer. The engineer or technician's resume must list applicable experience related to installation, start-up and testing of equipment and applicable factory training and education. Qualifications require the engineer to have supervised two installations of similar size and type which are operating satisfactorily. More than one engineer or technician may be provided based on the types of specific equipment. In event more than one engineer or technician is provided, provide a certified resume for each one and indicate his specific specialty and item of work on the resume.

1.6.9 Start-Up, Testing, and Installation Engineer

The Contractor must provide the services of a qualified engineer or technician for start-up, testing, and installation of equipment. The engineer or technician must be an employee of the equipment manufacturer.

1.6.10 Installation

Provide an installation engineer or technician to install and supervise

the installation of **steam boiler system** including instrumentation and boiler controls. Provide the technician or engineer until equipment installation is coordinated and completed with checkout.

1.6.11 **Start-Up Plan**

Submit a written schedule with dates of start-up, test, installation and checkout of equipment to the Contracting Officer.

1.6.12 **Start-Up and Test**

The specific equipment tested must be the boiler, boiler controls, boiler instrumentation and feedwater equipment. The start-up and testing engineer or technician must remain on the job until the unit has been successfully operated for [____] days and the unit has been accepted by the Contracting Officer.

1.6.13 **Start-Up Certification**

After installation of equipment, submit a certificate or certified written statement, signed by the engineer or technician, that the equipment is installed in accordance with the manufacturer's recommendations.

1.7 **DELIVERY, STORAGE, AND HANDLING**

Handle, store, and protect equipment and materials to prevent damage before and during installation in accordance with the manufacturer's recommendations, and as approved by the Contracting Officer. Replace damaged or defective items.

Protect equipment delivered and placed in storage from the weather, humidity and temperature variations, dirt and dust, and other contaminants.

1.8 **EXTRA MATERIALS**

Submit **spare parts** data for each different item of material and equipment specified, after approval of the **detail drawings** and no later than 2 months prior to the date of beneficial occupancy. Submit Detail Drawings consisting of equipment layout including installation details and electrical connection diagrams; combustion and safety control diagrams; ductwork layout showing the location of supports and hangers, typical hanger details, gauge reinforcement, reinforcement spacing rigidity classification, and static pressure and seal classifications; and piping layout showing the location of guides and anchors, the load imposed on each support or anchor (not required for radiant floor tubing), and typical support details. Include on the drawings any information required to demonstrate that the system has been coordinated and will properly function as a unit and to show equipment relationship to other parts of the work, including clearances required for operation and maintenance. Include in the data a complete list of parts and supplies, with current unit prices and source of supply, and a list of the parts recommended by the manufacturer to be replaced after 1 and 3 years of service.

1.9 **INSTRUCTION TO GOVERNMENT PERSONNEL**

When specified in other sections, furnish the services of competent instructors to give full instruction to the designated Government personnel in the adjustment, operation, and maintenance, including pertinent safety requirements, of the specified equipment or system.

Instructors must be thoroughly familiar with all parts of the installation and must be trained in operating theory as well as practical operation and maintenance work.

Instruction must be given during the first regular work week after the equipment or system has been accepted and turned over to the Government for regular operation. The number of man-days (8 hours per day) of instruction furnished must be as specified in the individual section. When more than 4 man-days of instruction are specified, use approximately half of the time for classroom instruction. Use other time for instruction with the equipment or system. When significant changes or modifications in the equipment or system are made under the terms of the contract, provide additional instruction to acquaint the operating personnel with the changes or modifications.

1.10 ACCESSIBILITY

NOTE: The following requirement is intended to solicit the installer's help in the prudent location of equipment when he has some control over locations. However, designer's should not rely on it at all since enforcing this requirement in the field would be difficult. Therefore, the system designer needs to layout and indicate the locations of equipment, control devices, and access doors so that most of the accessibility questions are resolved inexpensively during design.

Install all work so that parts requiring periodic inspection, operation, maintenance, and repair are readily accessible. Install concealed valves, expansion joints, controls, dampers, and equipment requiring access, in locations freely accessible through access doors.

1.11 WARRANTY

Materials and Workmanship Warranty: Pressure vessel must be warranted against defects in workmanship and materials for[1 year][2 years][5 years] after start-up or[1 year][2 years][5 years] from ship date, whichever is less.

Parts Warranty: parts and equipment must be warranted against defects in workmanship and materials for[1 year][2 years][5 years]<insert length of time after start-up.

PART 2 PRODUCTS

NOTE: In order to comply with UFC 1-200-02, designs must achieve energy consumption levels that are at least 30 percent below the ASHRAE 90.1 baseline. In accordance with P.L. 109-58 (Energy Policy Act of 2005), Executive Order 13423, and Federal Acquisition Regulation (FAR) Section 23.203 energy consuming products and systems must meet or exceed the performance criteria for ENERGY STAR®-qualified products as long as these requirements are nonproprietary. The ENERGY STAR product requirements

are available on the web at www.energystar.gov/products. Where ENERGY STAR products are not applicable, energy consuming products and systems must meet or exceed the requirements of ASHRAE 90.1.

ENERGY STAR Eligibility Criteria Version 3.0 requires that residential boilers of less than 90 kW 300,000 Btuh energy input supplying low pressure steam or hot water for space heating applications have a minimum AFUE of 90 percent for gas-fired type or 87 percent for oil-fired type. Boilers intended only for commercial applications and/or with an input rating of 90 kW 300,000 Btuh or higher are not eligible for Energy Star.

If a FEMP efficiency is not stated for a particular boiler capacity, this document includes boiler efficiency requirements in conformance with ASHRAE 90.1. ASHRAE 90.1 requires that low and medium pressure boilers used primarily in commercial space heating applications meet the following thermal (Et) or combustion (Ec) efficiencies:

Natural Gas-fired Hot Water rated at 88 - 732 kW 300,000 - 2,500,000 Btuh capacity, Et = 80 percent.

Natural Gas-fired Hot Water rated greater than 732 kW 2,500,000 Btuh capacity, Ec = 82 percent.

Natural Gas-fired Steam (excluding natural draft) rated at 90 kW 300,000 Btuh capacity and larger, Et = 79 percent.

Natural Gas-fired - Natural Draft Steam rated at 90 kW 300,000 Btuh capacity and larger, Et = 77 percent.

#2 Oil-fired Water rated at 88 - 732 kW 300,000 - 2,500,000 Btuh capacity, Et = 82 percent.

#2 Oil-fired Water rated greater than 732 kW 2,500,000 Btuh capacity, Ec = 84 percent.

#2 Oil-fired Steam rated greater than 90 kW 300,000 Btuh capacity and larger, Et = 81 percent.

Include all equipment efficiencies on the equipment schedules on the drawings.

2.1 MATERIALS AND EQUIPMENT

2.1.1 Standard Products

Provide materials and equipment which are the standard products of a manufacturer regularly engaged in the manufacture of the products and that essentially duplicate items that have been in satisfactory use for at least 2 years prior to bid opening. Equipment must be supported by a service organization that is, in the opinion of the Contracting Officer, reasonably convenient to the site. Submit manufacturer's catalog data

included with the detail drawings for the following:

- a. Radiant floor heating system including tubing, joints, and manifold for radiant floor heating systems.
- b. Data showing model, size, options, etc., that are intended for consideration. Data submitted must be adequate to demonstrate compliance with contract requirements. Data must include manufacturer's written installation instructions and manufacturer's recommendations for operation and maintenance clearances for the following:
 - (1) Boilers
 - (2) Unit Heaters
 - (3) Fuel Burning Equipment
 - (4) Combustion Control Equipment
 - (5) Pumps
 - (6) Fittings and Accessories
 - (7) Fuel Oil Storage System
 - (8) Water Treatment System
- c. Submit evidence that boilers meet requirements of standards specified. Include with the certificate of compliance acceptable evidence that standards are met. Acceptable evidence will be the official UL listing mark prescribed in the UL gas and oil equipment list for oil-fired, gas-fired, or gas and oil-fired boiler assemblies, as applicable plus the appropriate official ASME National Board, or for overseas installation a foreign nation symbol stamp. Any deviation from above requires approval from the cognizant Authority Having Jurisdiction (AHJ).

2.1.2 Asbestos Prohibition

Asbestos and asbestos-containing products will not be allowed.

2.1.3 Nameplates

Secure a plate to each major component of equipment containing the manufacturer's name, address, type or style, model or serial number, and catalog number. Also, display an **Energy Star** label as applicable. Each pressure vessel must have an approved ASME stamp.

2.1.4 Equipment Guards

Belts, pulleys, chains, gears, couplings, projecting setscrews, keys, and other rotating parts exposed to personnel contact must be fully enclosed or guarded in accordance with OSHA requirements. High temperature equipment and piping exposed to contact by personnel or where it creates a potential fire hazard must be properly guarded or covered with insulation of a type specified. Catwalks, operating platforms, ladders, and guardrails must be provided where shown and must be constructed in accordance with Section[**08 31 00** ACCESS DOORS AND PANELS][**05 51 33** METAL

LADDERS].

2.2 BOILERS

NOTE: Steam-atomizing burners should be specified
only if the boiler operating pressure exceeds **690**
kPa (gage) 100 psig.

NOTE: Oil burners can have turndown ratio of 4,5,
or 6 to 1.

NOTE: Repetitive self checking circuits are
required only on boilers with modulating controls
which operate for long periods of time without
burner shutdown.

NOTE: Steam operated feedwater injectors should be
specified only for boilers operating at or above **345**
kPa (gage) 50 psig.

NOTE: Combustion units should have metering
combustion controls. Investigate energy savings.
Combustion units larger than **7034 kW 24,000,000**
Btu/hr input should have oxygen trim in order to
optimize fuel usage. CO trim should be considered
for larger boiler installations as an adjunct to
oxygen trim for increased efficiency.

Provide boilers that conform to **ASME BPVC SEC I** and have a gross output capacity of at least [____] **kW Btu per hour** when operating at a steam pressure of [____] **kPa (gage) pounds per square inch gage (psig)** at the site under design conditions when the burner is firing No. [____] oil conforming to **ASTM D396** having a higher heating value of [____] **MJ/kg Btu per gallon** [or] [natural] [manufactured] [mixed] [liquified petroleum] gas having a higher heating value of [____] **MJ/kg Btu per cubic foot** and a pressure of [____] **kPa (gage) psig** at the fuel train connection]. Provide [air-atomizing] [or] [steam atomizing] burners. Rotary type burners are unacceptable. Provide burners capable of operating at a turn down ratio of [____]. Provide automatic [recycling] [nonrecycling] programming controls and incorporate means for automatic self-checking of the circuit beginning of each start-up cycle. [Include a repetitive self-checking circuit to check all components at not more than the specified time interval in **ASME CSD-1** during the entire period of burner operation, and sensor must be of the ultraviolet type.] [Connections must be provided for remote starting or stopping of the boilers.] [Provide steam operated feedwater injectors.] Provide combustion control system of the [modulating positioning type where the final control elements for fuel and air inputs, between minimum and maximum limits, are positioned in proportion to load] [semi-metering type where the final control element

for one input, either fuel or air, is positioned in proportion to load; flow of the other input is measured and controlled in proportion to load as indicated by the position of the control element for the first input][full metering (fuel flow/air flow basis) type where the final control element for one input, either fuel or air, is positioned in proportion to load, or the flow for that input is controlled in proportion to load, and the other input is measured and controlled in proportion to the measured flow of the first input].[Provide with oxygen compensation where one of the control elements is biased, modified slightly, on the basis of continuous flue gas analyses, to maintain the proper percentage of oxygen for various firing rates.] Boilers must be designed in accordance with the applicable provisions of **ASME BPVC SEC I** and current addenda therein. All control and safety devices must conform to **ASME CSD-1** for boilers with a fuel input rating of 400,000 btu/hr to 12,500,000 btu/hr or **NFPA 85** for boilers with a fuel input rating greater than 12,500,000. The following design requirements and conditions must apply:

- a. Maximum allowable steam working pressure of [_____] **kPa (gage) psig**, as designated by the symbol "P" in the ASME Code and determined by employing the allowable stress values, design rules, and dimensions therein.
- b. Operating pressure of [_____] **kPa (gage) psig** at the steam drum outlet.
- c. Feedwater temperature of [_____] **degrees C degrees F** at entrance to the boiler plant.
- d. Elevation of [_____] **meter feet** above sea level at the site.
- e. Ambient air temperatures outside the plant of [_____] **degrees C degrees F** minimum to [_____] **degrees C degrees F** maximum; average inside temperature of [_____] **degrees C degrees F** and relative humidity of [_____] percent.
- f. Steam quality of [99.5][99.0][_____] percent.
- [g. Make-up feedwater analysis of [_____] parts per million (ppm) total solids and [_____] ppm total alkalinity.
-]h. Steam temperature of [_____] **degrees C degrees F** at the steam drum outlet.

]2.2.1 Boiler Connections

Requirements for interconnecting piping, insulation, fuel supply,[vibration isolation] [_____] and other related work necessary to provide a complete and operable steam system, whether or not specifically mentioned above, must conform to applicable requirements of other sections of Division 15, and electrical work must conform to applicable requirements in Division 16.

2.2.2 Boiler Efficiency

[Gas fired boilers must have a steady state combustion efficiency of at least 80 percent.][Oil fired boilers must have a steady state combustion efficiency of at least 81 percent.]Boiler efficiency, when tested as specified herein, must be not less than[79-83] [percent when firing No. [_____] fuel oil][and][78-82][percent when firing [_____] gas]. Obtain efficiency at all loads in a range from the lower limit of the turndown

ratio to 100 percent.

2.2.3 Furnace Heat Input

When boiler is operating at maximum rated output, heat input to furnace must not exceed limits specified in **ASME BPVC SEC I** both for input per cubic meter foot of furnace volume and input per square meter foot of effective radiant heating surface.

2.2.4 Steam Quality

The steam releasing surface, drum internals, and steam passages out of the drum must be such that, with load swings from [50][75] to 100 percent of the rated capacity per [1][3] minute period at a boiler feedwater solids concentration of [3500 ppm][_____] ppm, the carryover must not exceed [_____] ppm at the boiler nozzle, and moisture in the steam must not exceed [0.5][1.0] percent.

2.2.5 Noise Levels

NOTE: The value for decibels for boiler must be that required for operating personnel and OSHA requirements taking into consideration room design and other requirements.

Conduct noise measurements and exposure analysis under the overall supervision of an industrial hygienist or suitably qualified medical officer from the Navy Regional Medical Center (NRMCC). Safety personnel, engineers and others who have been certified by the Chief, Bureau of Medicine and Surgery (BUMED) also may supervise the work. Maintain exposure limits for potentially hazardous noise levels of 85 dB, continuous or intermittent and 140 dB peak sound pressure, impulse or impact. Sound level meter must conform as a minimum to the Type 2 requirements cited in **ASA S1.13**.

2.2.6 Air Pollution Control

Note: Research local, state, and federal emission standards and place any unusual or new requirements in this specification section.

Provide fuel burning equipment, combustion control system(s) [and] stack(s) [and emission control equipment] designed, installed, and when applicable, adjusted so that emissions at any loading within the range of the boiler output specified do not exceed federal, state, and local environmental rules and regulations. Emission requirements to be considered include sulfur oxides, carbon monoxide, particulate, or nitrogen oxides. In addition, the density of any emission to the atmosphere, except during start-up, cleaning of fires, or soot blowing, must not exceed [No. 1][No. 2] on the Ringlemann Scale or the Smoke Inspection Guide. The plant must otherwise conform to applicable Federal, State and Local regulatory requirements. Air quality board "Permit to construct" [will be provided by the contracting officer][will be obtained by the contractor].

2.2.7 Firetube Boiler

Boiler must be self-contained, multipass, packaged type, complete with all accessories, mounted on a structural steel base. When the boilers are operating at maximum output, the heat input rates must not be greater than 21 Kw/square meter 6,700 Btuh per square ft of fireside heating surface.

2.2.8 Watertube Boiler

NOTE: Select between standard, finned, or bent/flexible tube boiler. If bent or flexible tube boilers are not selected, remove all references to bent or flexible tube boilers. Standard water tube boilers are steel tube boilers that have historically been used for most heating applications. Finned tube boilers are typically small boilers (residential type) that utilize a finned tube arrangement. Bent or flexible tube boilers are steel tube boilers with bent tubes that experience multiple water tube passes. Bent/Flexible tube boilers typically require less space than standard water tube boilers. The bent or flexible tubes are also easier to replace than the standard water tubes; however, the interior surface of the tubes cannot be cleaned mechanically; they can only be chemically cleaned. Bent or flexible tube boilers also have a higher ratio of heat output to heating surface area.

The boiler must be a[standard][finned][bent or flexible] type of water tube boiler. Boiler must be self-contained, packaged type, complete with all accessories, mounted on a structural steel base.[The boiler heating surface area for bent or flexible tube boilers must be at least 0.03 square meters/kW 4 square feet/boiler horse power.[The heat input rate for finned tube steam boiler or hot water generator must not be greater than 3.5 kW 12,000 Btuh based on internal heating area.] Bent or flexible tube boilers must be provided with single or multiple downcomers for circulation without the need for exterior pumping. The tubes for bent or flexible tube boilers must be designed for replacement without requiring welding or rolling of tubes. Any special tools required for bent or flexible tube removal or installation must be provided with the boiler.]

2.2.9 Cast Iron Boiler

Boiler must be of the rectangular, sectional type, self-contained, packaged type, complete with accessories, mounted on a structural steel base. Cast iron sections must be free of leaks under all operating conditions. Access must be provided to permit cleaning of internal tube surfaces.

2.2.10 Steam Heating Boilers

NOTE: Steam boilers will operate at pressures below 100 kPa 15 psi. In case of installation of a small boiler where the omission of the water column is standard in some manufacturers, the water column

requirement and other inapplicable words will be deleted. However, if the water column requirement is deleted from the specification, a visible water column must be included in the external piping arrangement to the boiler. The boiler feed water piping must contain a loop or trap. The bottom portion of the trap must be below the anticipated water level within the boiler. All piping arrangements will be shown on the drawings. Delete those subparagraphs which are not applicable to the design. Indicate the elevation of the project site and the outdoor ambient air temperature range expected at the project site. Site conditions affect fan selection, boiler design, and stack design. The feed water temperature must be selected to avoid thermal shock. Typical ranges are between 10 degrees C 20 degrees F and 20 degrees C 40 degrees F below the boiler outlet temperature. The boiler manufacturer should be consulted for proper selection. Select appropriate boiler types. Allow adequate space around each boiler to permit accessibility for operation, maintenance, and service (including space for tube removal). A minimum of 1200 mm 4 feet around the boiler will be required.

The boiler must be provided with a water column with gauge glass and fittings including water column and gauge glass drain valves of the straight through type. The steam heating boiler must be capable of operating at the specified maximum continuous capacity without damage or deterioration to the boiler, its setting, firing equipment, or auxiliaries. The rated capacity must be the capacity at which the boiler will operate continuously while maintaining at least the specified minimum efficiency. Design conditions must be as follows:

- a. Boiler design pressure 200 kPa 30 psig.
- b. Operating pressure at boiler outlet [_____] kPa psig.
- c. Steam temperature 120 degrees C 250 degrees F.
- d. Feedwater temperature [_____] degrees C degrees F.
- e. Outdoor ambient air temperature [_____] degrees C degrees F (max), [_____] degrees C degrees F (min).
- f. Site elevation [_____] m feet.
- g. Maximum continuous capacity [_____] kg pounds of steam per hour.
- h. Rated capacity [_____] kg pounds of steam per hour.
- i. Maximum exhaust stack temperature [_____] degrees C degrees F.

NOTE: Energy efficiency data for commercial boilers provided in item "j" below is from ASHRAE 90.1 Table 6.8.1F.

- j. [Gas fired boilers with a capacity less than 90 kW 300,000 Btuh must have an Annual Fuel Utilization Efficiency of at least 75 percent.][Oil fired boilers with a capacity less than 90 kW 300,000 Btuh must have an Annual Fuel Utilization Efficiency of at least 80 percent.][Gas fired boilers (all, except natural draft) with a capacity of greater than or equal to 90 kW 300,000 Btuh must have a thermal efficiency of at least 79 percent.][Gas fired natural draft boilers with a capacity greater than or equal to 733 90 kW 300,000 Btuh must have a thermal efficiency of at least 77 percent.][Oil fired boilers with a capacity greater than or equal to 90 kW 300,000 Btuh must have a thermal efficiency of at least 81 percent when fired at the maximum and minimum ratings allowed by the controls.]

2.3 BOILER CONSTRUCTION

2.3.1 General

Provide boilers constructed for specified design conditions and performance requirements in accordance with applicable provisions of ASME BPVC SEC I, and current addenda thereto, and ASME BPVC SEC I. Mount[the][each] boiler on an internal structural steel base frame of welded construction designed to serve as skids in shipment and to permanently support the unit off the prepared foundation at the site. Assemble and install tubes, drums, and headers in a manner to permit the boiler(s) to be drained dry by gravity.[Show location of the flue gas outlet on drawings.]

2.3.2 Drums

Drums must conform to ASME BPVC SEC I and requirements herein. Provide sufficient drum-internals to preclude thermal shock and ensure steam quality as specified herein. Ensure drum capacity addresses expected load swings to insure maintenance of drum level within set points during high load swings, and to properly condition the water in accordance with chemical feed and blowdown systems specified[indicated][and][specified]. Nuts, bolts, and other hardware used in assembling steam separators must be stainless steel. Where drum internals interfere with inspection of part of the boiler drum, make provisions for the easy removal and replacement of whatever internals must be removed in order to effect a complete internal drum inspection and to provide access to tubes. Construct lower drums to preclude build-up of suspended solids, and to permit the complete drainage.

2.3.3 Tubes

Boiler and furnace tubes must meet the applicable requirements of ASME BPVC SEC I. Tangent tubes or tubes with welded flat studs or a combination thereof must be provided for water cooling of the furnace rear wall, side walls, roof, and floor. Maximum tube spacing for tubes with welded flat studs should be given so that the problem of studs driving through tube sidewalls due to expansion and contraction does not occur. Burner wall must be refractory-covered with optional tubes for water cooling. Tubes must enter the steam drum below the normal water line.

2.4 BOILER TRIM

Boiler trim, including steam stop valves, feedwater stop and check valves,

blowoff valves, water level indicators, water columns, pressure gages, safety valves and associated piping, and fittings located within the jurisdictional limits of the ASME Code must conform to and be installed in accordance with the Code and must meet supplemental requirements specified herein. Material selection and pressure ratings for valves and piping must be in accordance with applicable ANSI standards cited in the ASME Code and must be based on the maximum allowable boiler working pressure specified herein except as otherwise permitted by the ASME Code.

2.5 FUEL BURNING EQUIPMENT

NOTE: Include all the required data for proper design of the boiler. Delete all references to fuels which will not be used. When firing fuel oil, include nitrogen and sulfur content of fuel for emission requirements.

Review the Clean Air Act Amendment of 1990 (CAAA) and other applicable Federal, state, and local regulations early in the design phase to determine the appropriate emission limitations and monitoring requirements.

The CAAA does not require the application of low NO_x burner (LNB) technology for boilers within the size range of this specification. The CAAA limits SO₂ emissions for fuel oil fired boilers over 10.55 gJ 10,000,000 Btu to 21.5 kg/nJ 0.5 lb per million Btu input or to firing oil with less than 0.5 weight percent sulfur. However, state implementation plans may place limits on NO_x and particulates and more stringent requirements on SO₂.

Many options are available to reduce NO_x emissions. The nitrogen and sulfur content of fuel oil should be specified in the fuel purchase contract. Restrictions on the nitrogen content will limit fuel flexibility. A careful analysis of proposed NO_x reduction technologies must be performed to account for any required changes to auxiliary equipment and to identify future increase in O&M costs. Important questions that should be answered and be a part of the evaluation include the performance of NO_x reduction over the entire load range, performance during backup fuel firing, and performance over the lifetime of the unit.

The majority of NO_x control techniques can be defined as combustion modifications. The goals of combustion modification include redistribution of air and fuel to slow mixing, reduction of O₂ in NO_x formation zones, and reduction of the amount of fuel burned at peak flame temperatures.

Combustion modifications primarily deal with the control of fuel and air. Vertical staging includes overfire air (OFA) ports above the main combustion zone. Horizontal staging use registers or other

devices to introduce air at different points along the flame. Fuel staging establishes a fuel rich zone above an air lean main combustion zone. Burner Out of Service (BOOS) techniques direct fuel to lower burner levels, while operating upper burners with air only. Flue Gas Recirculation (FGR) reduces O₂ available to react with nitrogen and cools the flame. In addition to low NO_x burners (LNB), OFA and BOOS other combustion modification techniques include fuel biasing, low excess air (LEA) and fuel reburning. Oil fired burners have successfully used advanced oil atomizers to reduce NO_x without increasing opacity. Oil/water emulsion is a technique to reduce NO_x on smaller industrial boilers.

Consideration will be given to the unique installation and space requirements of various NO_x reduction systems. LNB may or may not require pressure port modifications. FGR involves routing large ductwork. OFA is very effective and involves modification to pressure parts. Fuel staging requires pressure port modifications for reburn fuel injection and/or OFA ports.

Boiler must be designed to burn[gas][oil][combination gas and oil]. Each boiler must comply with Federal, state, and local emission regulations. As a minimum, the following emission requirements must be met:

NO_x -[[_____] kg/joule lb/million Btu input][parts per million (ppm) corrected to 3 percent O₂].

SO₂ -[[_____] kg/joule lb/million Btu input][parts per million (ppm) corrected to 3 percent O₂].

Particulate -[[_____] kg/joule lb/million Btu input][parts per million (ppm) corrected to 3 percent O₂].

2.5.1 Burners

NOTE: If No. 4, 5, or 6 oil will be one of the fuels, requirements for burners and accessories for these heavy oils will be inserted in the project specification.

2.5.1.1 Gas and Combination Gas-Oil Fired Burners and Controls

Burners must be UL approved[mechanical draft burners with all air necessary for combustion supplied by a blower where the operation is coordinated with the burner][natural draft/atmospheric burners]. Burner must be provided complete with fuel supply system in conformance with the following safety codes or standards:

- a. Gas-fired units with inputs greater than 0.117 MW 400,000 Btuh per combustion chamber must conform to UL 795.[Gas fired units less than

3.66 MW 12,500,000 Btuh input must conform to ANSI Z21.13/CSA 4.9.][
Single and multiple burner gas-fired units greater than or equal to
3.66 MW 12,500,000 Btuh input must conform to NFPA 85.]

- b. Combination gas and oil-fired units must conform to UL 296.[
Combination gas and oil-fired units less than 3.66 MW 12,500,000 Btuh
input must conform to ASME CSD-1.][Single and multiple burner
combination gas and oil-fired units equal to or greater than 3.66 MW
12,500,000 Btuh input must conform to NFPA 85.]

2.5.1.2 Oil-Fired Burners and Controls

Oil-fired burners and controls for oil-fired units firing No. [____] oil
must be atomizing, forced-draft type in conformance with UL 726.[
Oil-fired units less than 3.66 MW 12,500,000 Btuh input must conform to
ASME CSD-1.][Oil-fired units greater than or equal to 3.66 MW 12,500,000
Btuh input must conform to NFPA 85.]

[2.5.1.3 Steam or Air Atomizer

**NOTE: Delete this paragraph if steam or air
atomization is not used.**

[Steam][or][Air] atomizer must be of the inside mix type utilizing[
steam][or][air] mixing with the oil inside the nozzle. No moving parts
must be required within the atomizer assembly. Unit must be capable of
completely atomizing the oil through a minimum capacity range of 4 to 1
without changing nozzles or sprayer plates and when supplied with[steam][
or][air] at a maximum pressure of [100][____] kPa [15][____] psig.
Capacity of unit must be adjustable. Unit must be furnished with a
blowout valve so that[steam][or][air] may be blown through the oil
passages to clear them of any accumulation. A diffuser designed to
stabilize the flame must be mounted near the furnace end of the atomizer
in such a position that oil will not strike it.

]2.5.1.4 Mechanical Pressure Atomizer

**NOTE: If mechanical pressure atomization is not
used delete this paragraph.**

Mechanical pressure atomizer must operate solely by the use of oil
pressure and must have no moving parts within the atomizer. Unit must be
capable of completely atomizing the oil through a minimum capacity range
of 4 to 1 without changing nozzles or sprayer plates and when furnished
with oil at a constant pressure of [____]. A constant volume of oil must
be supplied to the atomizer. Variable capacity must be obtained by
adjusting control valve. A diffuser provided to stabilize the flame must
be mounted near the furnace end of the atomizer, but in such a position
that oil will not strike it.

]2.5.2 Draft Fans

**NOTE: If natural draft/atmospheric burners are
utilized, all draft fan paragraphs will be deleted.**

Select between forced draft and induced draft fan or a combination of both. In most applications, a forced draft fan will be adequate. Fan bearings on induced draft fans must have adequate means to prevent overheating and provision for lubrication. Choice of type of cooling will depend on availability of water for the particular site. Water-cooled bearings are generally used for induced draft fans but air-cooled, sealed-type bearings are available; however, their use must be approved by the fan manufacturer for the application if specified. Forced draft fans are typically air cooled.

Fans conforming to **AMCA 801** [forced-draft][and][induced-draft] must be furnished as an integral part of boiler design. Fans must be centrifugal with[backward-curved blades][radial-tip blades] or axial flow type. Each fan must be sized for output volume and static pressure rating sufficient for pressure losses, excess air requirements at the burner, leakages, temperature, and elevation corrections for worst ambient conditions, all at full combustion to meet net-rated output at normal firing conditions, plus an overall excess air volume of 10 percent against a 20 percent static overpressure. Noise levels for fans must not exceed 85 decibels in any octave band at a **0.914 m 3 foot** station.[Forced draft fan bearings must be air cooled.][Induced-draft fans must be designed for handling hot flue gas at the maximum outlet temperature in the boiler. Induced draft fan housings must be provided with drain holes to accommodate the drainage of condensation. Induced draft fan bearings must be[air-cooled][water-cooled]. Induced draft fan scroll sheets and rotor blades must have protective liners.]

2.5.2.1 Draft Fan Control

[Forced-draft centrifugal fans must have inlet vane controls or must have variable speed control where indicated. Inlet vanes must be suitable for use with combustion control equipment.][Induced-draft centrifugal fans must have outlet dampers and must have variable speed control.][Induced-draft fans must have inlet vane controls.]Axial propeller fans must have variable propeller pitch control.

2.5.2.2 Draft Fan Drives

NOTE: Where motor starters for mechanical equipment are provided in motor control centers, delete the description of motor starters.

Fans must be driven by electric motors. Electric motor must be[drip proof][totally enclosed nonventilated][totally enclosed fan cooled][totally enclosed fan-cooled, suitable for installation in a Class II, Division 1, Group F, hazardous location conforming to **NFPA 70**].[Motor starter must be[magnetic across-the-line][reduced voltage start] type with[general purpose][weather-resistant][watertight][dust-tight][explosion-proof] enclosure and must be furnished with four auxiliary interlock contacts.]

]2.5.3 Draft Damper

NOTE: Select between manual and automatic dampers.
Normally, manual dampers are adequate for single
boilers less than 600 kW 2,000,000 Btuh capacity.
Select automatic dampers for modular boilers.

Boilers must be provided with[manual][automatic] dampers, draft hoods, or barometric dampers as recommended by the boiler manufacturer to maintain proper draft in the boiler. Draft damper must be provided in a convenient and accessible location in the flue gas outlet from the boiler. Automatic damper must be arranged for automatic operation by means of a[damper regulator][furnace draft regulator][damper motor].

]2.5.4 Ductwork

NOTE: In colder climates, tempering of combustion
air may be required. Add an appropriate paragraph
for tempering combustion air, if required. Delete
this paragraph, if a plenum chamber is not needed.

Air ducts connecting the forced-draft fan units with the plenum chamber must be designed to convey air with a minimum of pressure loss due to friction. Ductwork must be galvanized sheet metal conforming to **ASTM A653/A653M**. Ducts must be straight and smooth on the inside with laps made in direction of air flow. Ducts must have cross-break with enough center height to assure rigidity in the duct section, must be angle iron braced, and must be completely free of vibration. Access and inspection doors must be provided as indicated and required, with a minimum of one in each section between dampers or items of equipment. Ducts must be constructed with long radius elbows having a centerline radius 1-1/2 times the duct width, or where the space does not permit the use of long radius elbows, short radius or square elbows with factory-fabricated turning vanes may be used. Duct joints must be substantially airtight and must have adequate strength for the service, with **38 x 38 x 3 mm 1-1/2 x 1-1/2 x 1/8 inch** angles used where required for strength or rigidity. Duct wall thickness must be 16 gauge (**1.5 mm**) (**0.0598 inch**) for ducts **1500 mm 60 inches** or less and 12 gauge (**2.66 mm**) (**0.1046 inch**) for ducts larger than **1500 mm 60 inches** in maximum dimension. Additional ductwork must be in accordance with Section **23 30 00**
HVAC AIR DISTRIBUTION.

]2.6 COMBUSTION CONTROL EQUIPMENT

Controls for facilities with operating Energy
Monitoring and Control Systems (EMCS) will be
specified to be compatible with existing EMCS
controls. Delete reference to multiple boilers if a
single boiler is used.

Delete "pneumatic controls" for new systems. Use
only pneumatic controls for existing pneumatic
controls systems that require new parts be purchased.

Combustion control equipment must be provided as a system by a single manufacturer. Field installed automatic combustion control system must be installed in accordance with the manufacturer's recommendations and under the direct supervision of a representative of the control manufacturer. [The boiler pressure must be controlled by a steam pressure controller.] The equipment must operate [electronically] [either electrically or pneumatically as applicable]. On multiple boiler installations, each boiler unit must have a completely independent system of controls responding to the load and to a plant master controller. If recording instruments are provided, a 1 year supply of ink and 400 blank charts for each recorder must be furnished.

[2.6.1 Pneumatic Controls

If pneumatic operation is provided, a regenerant desiccant air dryer unit must be provided. Boiler must shut down on loss of control air pressure. Pneumatic control systems must conform to CAGI B19.1. Air filter regulator sets must be installed at each control valve and transmitter in the system. The master air filter regulator set on the control panel must be the dual type where one side can be cleaned and repaired while the other is operating. Exterior control air piping and devices must be protected from freezing.

2.6.1.1 Air Compressor Unit

The air compressor unit must be electric-motor driven, polytetrafluoroethylene or carbon ring type automatic air compressor. The compressor unit must be sized to run not more than 60 percent of the time when all controls are in service. The air compressor unit must be complete with necessary accessories including automatic pressure control equipment, relief valves, check valves, air filters, moisture traps, and a receiver with ample capacity for emergency operation of the controls for 15 minutes after compressor shutdown. Compressor speed must not exceed 900 rpm. Motor speed must not exceed 1750 rpm. The compressor air intake must be provided with a low drop type air suction filter/silencer suitable for outdoor installation.

2.6.1.2 Air Receiver

**NOTE: The condensate drain line will be located in
such a manner as to prevent freezing.**

The air receiver must be constructed in accordance with ASME BPVC SEC VIII D1 for unfired pressure vessels for 1379 kPa 200 psi working pressure, and must be equipped with inlet and outlet connections, valved drain connection, minimum 150 mm 6 inch dial pressure gauge, pop safety valves, and regulator connections.

]2.6.2 Electrical controls

Electrical control devices must be rated at [120][24] volts and must be connected as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

2.6.3 Steam Pressure Controller

The controller must be of sturdy construction and must be protected

against dust and dampness. The sensing elements of the steam controller must be in direct contact with the steam.[Fixed position (on-off) and three position (high-low-off) type controllers must operate on a 6.9 kPa 1 pound differential over a pressure range of 0 to 103.4 kPa 0 to 15 psig.][Modulating controllers must automatically maintain, within 2 percent, the desired steam pressure by regulating the burner.]

[2.6.4 Boiler Plant Master Controller

NOTE: If only one boiler is utilized, the following paragraph will be deleted. A master controller will be provided for applications involving multiple boilers and for boilers arranged in a modular configuration.

A boiler plant master controller, sensitive to a[temperature transmitter in the return water header for the boiler][steam pressure transmitter in the boiler steam discharge header] must be furnished to provide anticipatory signals to all boiler controllers. Boiler controllers must react to anticipatory signals from the plant master controller as necessary in response to the boiler[temperature][pressure] indication to maintain the preset[temperature][pressure]. An automatic-manual switch must be provided to allow the sequence of boiler loading to be varied to distribute equal firing time on all boilers in the plant. The plant master controller must load the boilers one at a time as the plant load increases.

]2.6.5 Boiler Combustion Controls and Positioners

NOTE: A pilot is required for all oil fired boilers over 875 kW 3,000,000 Btuh. However, a pilot is recommended for all fired boilers.

Select between fixed rate (on-off), three position (high-low-off), and modulating controls. Combustion controls will be fixed-rate, on-off for gross outputs up to 200 kW 700,000 Btuh; high-low-off or fixed-rate on-off, depending on anticipated load profile, for gross output from 200 to 600 kW 700,000 to 2,000,000 Btuh; high-low-off or modulating, depending on anticipated load profile, for gross output from 600 to 1200 kW 2,000,000 to 4,000,000 Btuh; modulating for gross outputs above 1200 kW 4,000,000 Btuh. Modular boilers will be fixed-rate on-off for each module. Delete inappropriate paragraphs.

- a. [Gas][Combination gas-oil fired]boiler units must be provided with[fixed rate (on-off)][three position (high-low-off)][modulating] combustion controls with gas pilot or spark ignition. Modulating controls must be provided with a means for manually controlling the firing rate.
- b. Oil fired boiler units must be provided with[on-off][high-low-off][modulating] combustion controls with[direct electric spark ignition

system][spark ignited[No. 2 oil][natural gas][liquified petroleum gas] pilot]. Modulating controls must be provided with a means for manually controlling the firing rate.

- c. Modulating control function must be accomplished using positioning type controls. Air flow ratio and fuel control valve must be controlled by relative positions of operative levers on a jackshaft responding to a steam pressure controller. Positioning type combustion control equipment must include draft controls with synchronized fuel feed and combustion air supply controls, while and must maintain the proper air/fuel ratio. The desired furnace draft must be maintained within 0.25 mm 0.01 inch of water column.
- d. [Fixed rate on-off][High-low-off]controls for boilers with capacities up to 600 kW 2,000,000 Btuh must use a[temperature well in direct contact with the water][steam pressure controller in direct contact with the steam].

2.6.6 Combustion Safety Controls and Equipment

NOTE: Provide feed water regulator with low-water cutoff on close coupled boilers (i.e., short supply and return lines with low pick-up losses) under 1200 kW 4,000,000 Btuh gross output with no process loads. Provide pump controller with low-water cutoff for all other boilers. Low-water cutoff will require a manual reset unless a supplementary low-water cutoff is provided. A supplementary low-water cutoff is required for boilers utilizing a pumped condensate return system. State and local codes may also require supplementary low-water cutoffs. When a supplementary low-water cutoff is provided, it will require manual reset and the initial low-water cutoff will not require manual reset.

Include the manually operated shutoff switch in the controls drawings, set point schedules and plans.

Combustion safety controls and equipment must be UL listed, microprocessor-based distributed process controller. The system must include mounting hardware, wiring and cables, and associated equipment. The controller must be mounted completely wired, programmed, debugged, and tested to perform all of its functions. The controller must process the signals for complete control and monitoring of the boiler. This must include maintaining boiler status, starting and stopping all control functions, sequencing control functions and signaling alarm conditions. The program must be documented and include cross references in description of coils and contacts. Microprocessor must be able to perform self diagnostics and contain a message center to provide operator with status and failure mode information. Controllers for each boiler must be mounted on a separate, free standing panel adjacent to the boiler or for packaged boilers on the boiler supporting structure. Control systems and safety devices for automatically fired boilers must conform to ASME CSD-1. Electrical combustion and safety controls must be rated at 120 volts, single phase, 60 Hz and must be connected as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. A 100 mm 4 inch diameter alarm bell must be

provided and must be located where indicated or directed. The alarm bell must ring when the boiler is shut down by any safety control or interlock. Indicating lights must be provided on the control panel. A red light must indicate flame failure, and a green light must indicate that the main fuel valve is open. The following shutdown conditions must require a manual reset before the boiler can automatically recycle:

- a. Flame failure.
- b. Failure to establish pilot flame.
- c. Failure to establish main flame.
- d. [Low-water][Supplementary low-water]cutoff.
- e. [High temperature cutoff][High pressure cutoff].

2.6.6.1 Low-water Cutoff

Low water cutoff must be float actuated switch or electrically actuated probe type low-water cutoff. Float chamber must be provided with a blow-down connection. Cutoff must cause a safety shutdown and sound an alarm when the boiler water level drops below a safe minimum level. A safety shutdown due to low water must require manual reset before operation can be resumed and must prevent recycling of the burner. The cutoff must be in strict accordance to [ASME CSD-1](#).

2.6.6.1.1 Feedwater Regulator with Low-water Cutoff

Regulator must be an approved design sized for the application. A regulator must be provided for each boiler. The feeder must be so arranged that water will be fed to the boiler automatically when the water level in the boiler drops below a preset point and will actuate the alarm bell when the water level reaches the low danger point. The boiler feeder must be arranged so that the burner and forced-draft fan will stop whenever the water level drops below a preset danger point. The boiler feeder must be constructed so that the feedwater valve and seat are isolated from the float chamber to prevent overheating of the feed water and precipitation of scale on either the valve or seat. Each float mechanism, valve, and seat must be constructed of an approved, durable, corrosion-resistant steel alloy. Valve seats must be removable and renewable. The regulator must be equipped with a large, self-cleaning strainer. The drain valve on the regulator must be the gate or other straight-through type.

2.6.6.1.2 Pump Controller with Low-water Cutoff

Controller must be a design approved by the boiler manufacturer. A pump controller must be provided for each boiler which is used for space heating and process steam loads or long distribution lines. Pump controller must control the operation of the burner, forced-draft fan, and pump. Pump controller and low-water cutoff must have a float-operated[mercury][non-mercury] switch arranged to start and stop the pump at preset boiler water levels. If the water level in the boiler reaches the low danger point, a second switch must shut down the burner and actuate the alarm bell.

2.6.6.1.3 Supplementary Low-water Cutoff

Supplementary low-water cutoff of the[electrically operated probe type][float activated type] must be provided in addition to the low-water cutoff required above on each boiler. Supplementary low-water cutoff must be mounted directly in the boiler shell and must be set below the low-water cutoff required above.

2.6.6.2 Water Flow Interlock

Hot water boiler limit controls must be provided to include protection for low boiler water flow and high boiler water temperature. The limit controls must be interlocked with the combustion control system to effect boiler alarm and shutdown. The controls must not allow boiler startup unless hot water flow is proven.

2.7 BOILER INSTRUMENTATION

In addition to the instruments required by the boiler specification referenced above, provide the following instruments:

- a. Temperature gages to indicate temperature of[preheated oil][air preheater inlet and outlet air][steam temperature].
- b. Meters for indicating air flow, drum level and indicating, recording, and totalizing steam and fuel flow. Installation and calibration of these meters must confirm to ASME PTC 19.5.
- c. Instruments to indicate and record flue gas temperature.
- d. A draft gage, diaphragm or bellows type, conforming to ASME B40.100. Circular scale with graduations in Pa inches of water. Numerals must be suitable for reading by persons with normal vision from a distance of 6 meters 20 feet. Select range for intended service and contain completely within the calibrated range. Encompassing no more than 80 percent of the total range. Gages must be designed and constructed to ensure accuracy within plus or minus 2 percent of the full scale reading in ambient temperatures varying from 2 degrees C to 57 degrees C 35 degrees F to 135 degrees F. Operating mechanisms must withstand, without affecting gage accuracy or damage to the instrument, surges equal to 100 percent above or below the entire scale range. Gages must be calibrated prior to installation and must require no further calibration or adjustment other than setting the pointer to zero. Provide gages with a single pressure connection to the diaphragm, or bellows. Provide pressure connections with a 3-way stopcock which will permit checking of the zero setting and calibration at any time without breaking service connections.

NOTE: Select the applicable paragraph(s) from the following

- [e. A carbon dioxide recorder which must measure, record, and indicate the percentage by volume of carbon dioxide (CO₂) detected in the flue gas. The recording unit must be flush mounted and furnished with locking device and master key.
-][f. Provide complete oxygen analyzer system to measure oxygen content of

flue gases generated by combustion of[gas][or][oil] as specified in paragraph BOILER PLANT CONTROLS AND INSTRUMENTS herein for each boiler. The output of the analyzer must range from zero to 25 percent oxygen. The analyzer must include the following:

- (1) Provide complete aspirating system with proper connection to stack, necessary steam or water aspirating facilities, and piping of proper specification to analyzer. Provide piping in accordance with the oxygen analyzer manufacturer's recommendations and install tight. Install equipment in accordance with manufacturer's instructions.
- (2) Provide a paramagnetic analyzer. Analyzer must provide oxygen analysis in the zero to 25 percent oxygen range, and have means of calibration. Provide zero range and span adjustments as required.
- (3) Analyzer output and recorder input must be compatible. Provide necessary transducers, connect, and place in proper operation. Follow special instructions relating to electrical transmission between analyzer and recorder as to the application of shielded wiring in conduit.
- (4) Check system with two calibrating gases as follows: (a) 100 percent nitrogen, and (b) 3.5 percent oxygen and remainder nitrogen.

][g. Direct Probe In Situ Type: Oxygen analyzer must be the direct probe type utilizing an in-situ zirconium sensing element. Determine flow characteristic and insert element in location where flue gas is in transition and is representative. Sensing element must be contained within a protective shield mounted to the ductwork by means of an adapter plate, all furnished by the manufacturer. Analyzer must be equipped with a facility to allow daily automatic calibration check without removing the analyzer from the process. That is, sample gases may be injected directly on the sensing element while the analyzer is in the process. In order to eliminate temperature effect of flue gases, maintain the cell temperature in the probe at 843 degrees C 1,550 degrees F by means of an externally mounted temperature controller equipped with cold junction compensation and coupled to the probe with at least 6 meters 20 feet of flexible cable. Analyzer must be FM APP GUIDE approved and certified for "in-stack" analysis technique. Output signal range must be 4 to 20 milliamps and must represent 0.25 percent to 25 percent oxygen as a logarithmic function, 0.1 percent to 10 percent oxygen as a logarithmic function, or zero percent to 10 percent as a linear function.[Circular chart recorder must consist of a two-pen recording control mechanism having 110 volt a.c. electric motor drive. Chart must be 300 mm 12 inch diameter and rotate once every 24 hours. Output control signal will be 20 to 103 kPa (gage) 3 to 15 psig pneumatic. Sufficient blank charts and four ink cartridges per pen for 400 days operation must be provided. Recorder must have a dual 30 degrees strip indicator. Strip indicator and chart paper scale must be logarithmic or linear and consistent with the analyzer signal conditioning.][Strip chart recorder must consist of a two pen solid state electronic recording/controlling mechanism. Strip chart must be 100 mm 4 inches wide and must be driven at a speed of 19 mm per hour 3/4 inch per hour. Recording/controlling will operate on 110 volt ac power. Recorder must be furnished with twelve usable 24 hour logarithmic or linear

charts consistent with the analyzer signal conditioning. Inking system must be a breakaway inking system with replaceable fiber tip pens and 12 ink cartridges. Strip chart recorder/controller must have vertical scale and horizontal driven chart. Output control signal will be 4 to 20 milliamps dc.] Flue gas temperature scale must be 149 to 427 degrees C 300 to 800 degrees F. Entire system response must be not more than 3 seconds.

-] h. A smoke density[indicator][recorder] of the[density limit][continuous density] type with a scale calibrated in Ringlemann units. Indicating and recording systems must include circuits for the audible warning of the maximum smoke density-limit. A vibrating electric horn must be supplied to sound the audible signal.
- [i. The annunciator must be of the solid state type and must annunciate safety interlock and limit switch shut-downs, including flame failure, high and low water, high steam pressure,[high gas pressure,][low gas pressure,][high oil pressure,][low oil pressure,][low atomizing steam pressure,][draft fan failure,][high and low temperature,][and] [____]. Annunciator must include at least [two][____] spare points and audible alarms for at least high and low water conditions. Annunciator must not require special or additional devices which could affect the sensitivity of the boiler limit switches. Annunciator must provide means to indicate the first limit or off-limit condition causing burner shutdown and to sound an alarm. Annunciator must have a test button and a feature capability to store information on additional causes of burner shutdown. Subsequent causes of burner shutdown will be indicated after the operator acknowledges the first burner shutdown. A dust tight enclosure must be provided.

]2.7.1 Instrumentation

[Instrumentation, such as the steam flow recorder, steam flow totalizer, air flow recorder,[carbon dioxide][oxygen] recorder, and flue gas temperature recorder may be combined into one or more meters having multiple pens plus integrating capabilities as required for specified totalization functions.][Instrumentation for recording and integrating steam flow, for recording percent of CO₂ or O₂ in the flue gas, and for recording flue gas temperature must be combined in a single boiler operation meter having three pens using different colored ink.]

2.7.2 Control Panel

The boiler control panel must include the following switches, controls, and accessories:

- a. Boiler on-off pushbutton station with indicating lights.
- b. Draft fan manual, start-stop, pushbutton station with indicating lights for on, off, and tripped.
- c. Controllers for the combustion control system.
- d. Manual-automatic stations for individual controllers of the combustion control system.
- e. [Indicating lights and alarm with silencer switch to signal nonrecycling safety shutdowns as specified herein[and excess smoke density conditions].][Annunciator.]

- f. Manual switch to transfer combustion burner and controls from oil to gas and vice versa.
- g. Switch and indicating lights to control emergency gas shut-off valve.
- h. [Electric clock, 300 mm 12 inch dial, with second sweep hand and seal-in driving unit.][Digital clock that indicates hour, minutes, and seconds, with numerals suitable for reading by persons with normal vision from a distance of 6 meters 20 feet.]

2.7.3 Control Systems

The flame safeguard and programming control systems must be mounted in the central control panel or an auxiliary control panel located adjacent to the main panel adjacent to the boiler. Auxiliary panels, if provided, must be fabricated as specified herein for the main control panel.

2.8 BOILER PLANT CONTROLS AND INSTRUMENTS

Provide the following plant[controls and] instruments:

- a. Instruments to indicate and record feedwater temperature and steam pressure in the deaerator.
- b. Pressure gages to indicate steam pressure in main header, conforming to ASME B40.100[gas pressure in main supply line][oil pressure][atomizing steam pressure].
- c. A flue gas analyzer, Orsat type. The analyzer must determine the CO₂, CO, O₂ in the flue gas and must be complete with chemicals and accessories for use in such determinations.
- d. A steam flow recorder which must remotely indicate, record, and totalize the steam flow per hour through the steam header. Provide panel-mounted indicating recorder with a tamper-proof case.

NOTE: Insert section numbers for oil and/or gas piping systems in the blanks below.

- e. Volumetric fuel flow meters in accordance with [____],[and] [____].

NOTE: Master combustion controller should be used only for automatic recycling boilers.

- f. A common boiler master combustion controller located on the free standing instrument and control panel to control all boilers with each individual boiler controller acting as a submaster controller. The boiler master control system must provide for base loading one[or more] boilers. Base loaded boilers must be selected manually by an externally accessible switch. On call for heat, lead boiler must cut in and moderate firing rate to satisfy demands. When maximum desired firing rate is reached, lag boiler or boilers must cut in. Only one boiler must be on modulating firing at one time. Maximum desired firing rate for base loaded boiler must be adjusted initially for

boiler peak efficiency and must be capable of easy manual adjustment by operating engineer. Adequate indicators must be provided to show the method of loading of each boiler, and load being carried by it. Make adjustments at the front panel and no linkage adjustment must be necessary.

- g. The combustion control system must be capable of maintaining the plant steam pressure at the main header within the tolerance limits of [_____] percent expressed as a percent of the set point values. The specified tolerance must apply to any load which, within a one minute period, swings from a steady-state condition to an increase (or decrease) in load equal to a maximum of 10 percent of the plant. Regulation tolerances must apply to any steady state condition within the plant turndown ratio of [_____] . After a load swing of 10 percent the concentration of oxygen in the flue gas must not vary more than plus or minus [_____] percent of the values established at time of system set up within a time period of 30 seconds for firing rates of or above 33 percent of maximum, and plus or minus [_____] for firing rates below 33 percent of maximum.[For oxygen compensation, a continuous analysis of flue gases leaving the boiler must be made and recorded by an oxygen indicator and recorder mounted on the control panel. From this analysis, a signal must be sent to the fuel-air ratio system which must be biased in response to the signal to maintain the oxygen levels within plus or minus [10][_____] percent of the original setting for excess combustion air]. Combustion efficiency must not be less than that specified in the boiler specification.

2.9 BOILER CONTROL AND INSTRUMENT CABINET(S)

Provide boiler control and instrument cabinet(s) as specified in ASME BPVC SEC I, and may be mounted either on the boiler front or adjacent thereto. The arrangement may also consist of a boiler mounted cabinet containing controls normally provided by the manufacturer and a supplementary cabinet containing controls normally provided by the manufacturer and a supplementary cabinet containing additional controls and instruments required herein.[[Plant master combustion control][and][Steam flow recorder] must be mounted on or be adjacent to control panel for number [_____] boiler.]

2.10 FREE-STANDING MULTI-BOILER PLANT CONTROL AND INSTRUMENT PANEL

Provide a free-standing panel and locate as indicated. The panel must contain all multi-boiler controls, monitoring systems, and panel mounted instruments specified herein and in the reference specifications, except that flame safeguard systems may remain separately mounted in a cabinet at each boiler.

2.10.1 Control Panel Construction

Control panel must be constructed of not less than 11 gage (3 mm) (0.1196 inch) reinforced steel for face, top, and sides. The enclosed panel must be not less than 610 mm 24 inches in depth with inside rigidly welded braces. Control panel must be designed so that all indicating and recording devices and manually operated switches must be flush mounted in a gasketed removable top front panel with indicating and recording devices at eye-level. A similar removable top rear panel located opposite front panel must be provided to facilitate wiring, piping, and maintenance. Other operating controls must be installed on a sub-panel within the

enclosure. Access to panel enclosure must be through gasketed, double piano-hinged doors of not less than 16 gage (1.52 mm) (0.0598 inch) steel. Reinforced doors to prevent sagging and provided with a 3 point compression type fastener and polished key lock handle. A full width fluorescent lighting canopy must be prime coated and finished in baked enamel. Identify flush mounted devices with engraved lamcore nameplates. Adequately reinforce, suitably skirt, and suitably design panel base to permit anchoring to the floor or foundation.

2.10.2 Control Panel Wiring and Piping

Control panel must be factory pre-wired in accordance with NFPA 70. Wire must be thermoplastic Type THW, THWN, XHHW, or UL approved for the intended use, color or number coded, and run in plastic ducts to numbered terminal blocks. Control circuits must be separately fused with properly rated cartridge type fuses. Power leads to and from magnetic starters and contactors must terminate at terminal blocks so that no field wiring in control compartment is necessary except from terminal blocks to external equipment. Control leads to and from external control devices must terminate at separate terminal blocks from power leads. Steam, draft, and air operated devices must be factory piped to permanently affixed external connections. Pneumatic signals must be either 20 to 103 kPa (gage) 3 to 15 psig or 20 to 207 kPa (gage) 3 to 30 psig. Piping connections to indicators must be copper tubing conforming to ASTM B88M ASTM B88. The boiler operating switch must have a dust-tight sealed snap-action type. The precision switches must have cadmium, silver, or platinum contacts, wiping action type, rated at 10 amperes. Electrically[or] pneumatically test controls and equipment to simulate complete operational sequence.

2.11 BREECHING

Connect breeching to[the][each] boiler and to intermediate heat recovery equipment as required. Structural materials must comply with the applicable sections of the AISC 360, "Structural Steel Buildings Allowable Stress Design and Plastic Design." Shop connections must be welded gas-tight. Field connections may be welded or bolted as required for joining breeching to equipment. Hot dipped galvanized bolts and lock washers must be supplied for all bolted connections, must not be less than 10 mm 3/8 inch diameter and spaced not more than 76 mm 3 inches apart. Furnish bolted joints with 3.20 mm 1/8 inch thick non-asbestos gaskets. Breeching plate must be not less than 10 gage (3.40 mm) (0.1345 inch) in thickness. Welding must conform to AWS Z49.1 and AWS D1.1/D1.1M. Breeching system must provide for maximum expansion and contraction. Expansion joints must be of the guided, flexible-crease type with flexible element of not less than 16 gage (1.50 mm) (0.0598 inch) thick stainless steel. Provide access doors constructed of cast iron or reinforced steel plate with gaskets and positive closing latches of sufficient number to ensure a gas-tight seal. Clean breeching of rust and scale after fabrication. Insulation and covering for breeching must be as specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

2.11.1 Round Breeching

Construct round breeching 305 mm 12 inches in diameter and smaller of not less than No. 10 gage (3.40 mm) (0.1345 inch) steel with longitudinal groove type seam. Joints between lengths of breeching must be of the tight slip fit type with not less than 100 mm 4 inch engagement. Construct round breeching over 305 mm 12 inches in diameter of not less than No. 10

gage (3.40 mm) (0.1345 inch) steel with welded seams and joints. Construct round breeching over 457 mm 18 inches in diameter of not less than No. 10 gage (3.40 mm) (0.1345 inch) tank steel with seams and joints welded.

2.11.2 Expansion Joints

2.11.2.1 Metallic Breeching Expansion Joints

Provide factory fabricated metallic breeching expansion joints[where indicated]. Expansion joints must be guided metal bellows type capable of a minimum of [_____] mm inches of axial travel. Form metal bellows from not less than 1.60 mm 1/16 inch thick type 321 stainless steel plate. Cover plates must be not less than 3.20 mm 1/8 inch thick steel plate.

2.11.2.2 Non-Metallic Expansion Joints

Provide factory fabricated non-metallic breeching expansion joints 3.20 mm 1/8 inch minimum thickness[where indicated]. Expansion joints must be constructed of a fluoroelastomer vulcanized to two plies of knitted wire mesh capable of a minimum of [_____] mm inches of axial compression, [_____] mm inches of axial extension and [_____] mm inches of lateral offset[unless indicated otherwise]. Joints must have a continuous operating temperature rating of 204 degrees C 400 degrees F, with excursion design standards up to 400 degrees C 750 degrees F. Operating pressure range must be minus 34 kPa (gage) 5 psig to plus 34 kPa (gage) 5 psig. Expansion joints must be preformed with integrally molded corners, suitable for mounting against a 150 mm 6 inch flange.[Provide carbon steel backup bars with slotted holes, bolts, and nuts.]

2.11.3 Breeching Hangers

Design breeching hangers to carry not less than five times the breeching weight. Hangers for round breeching must be of the band type, with hanger rods. Hangers for rectangular breeching must be of the trapeze type, with angle support member and hanger rods. Make all hangers of steel.

2.11.4 Cleanout Doors

Secure cleanout doors to the ends and sides of the breeching where indicated or where required to effectively clean the breeching. Construct cleanout doors of a gage steel not less than that of the breeching and secure to a 32 by 32 mm 1 1/4 by 1 1/4 inch angle frame and spaced not over 152 mm 6 inches on center; provide 1.60 mm 1/16 inch thick fiber non-asbestos gasket between cleanout doors and frames. Doors must be squared and must be full height of diameter or side of breeching up to a size of 610 mm 24 inches maximum, except for cleanout doors less than 305 mm 12 inches in height must be rectangular and must be 305 mm 12 inches in length. Plug type cleanouts are not acceptable. Provide additional cleanout doors where needed.

2.12 BOILER STACKS

Provide [one][_____] [refractory-lined],[dual wall insulated,] steel stack(s), [each] [_____] mm inches in diameter inside [refractory] by [_____] meter feet overall length, and install where indicated.

2.12.1 Construction

[The][Each]stack may be factory or field assembled. [The][Each]stack must be[prefabricated double or triple walled][fabricated] of 6.40 mm 1/4 inch structural steel plates in accordance with ASTM A283/A283M. Stack sections must be of welded construction and fabricated in sections not greater than 6 meters 20 feet nor less than 1.50 meters 5 feet in length. Longitudinal seams must have continuous butt welded joints. The section or horizontal joints of [refractory lined] stacks must be flanged with angles welded to the shell, and sections bolted together. Angles must be not less than 50 mm by 50 mm by 6.40 mm 2 inches by 2 inches by 1/4 inch.[The refractory joint must be "buttered" with a high temperature air setting bonding refractory before each section is assembled.] Stacks must be set on a 13 mm 1/2 inch thick steel base plate having a diameter 50 mm 2 inches larger than the outside diameter of the base angle. Base angles and anchor must be a minimum of 10 mm 3/8 inch thick and 13 mm 1/2 inch diameter. A reinforced concrete foundation must be provided. Concrete for foundation and reinforcing must be as detailed on drawings and as specified under the concrete section of this specification. Provide stacks with steel guide shoes and bearing plates on all four sides at the roof level. Guides and bearing plates must be structural shapes not less than 6.40 mm 1/4 inch thick. Guides must be not less than 305 mm 12 inches long and must be welded to the stacks. Bearing member must be fastened with steel stud anchors welded to the bearing member. Provided a clearance of 3.20 mm 1/8 inch between guide and bearing member for stack diameter expansion after stacks are in normal operation. Provide flashing and counter-flashing around[the][each] stack as detailed on drawings.

2.12.2 Breeching Connection

Provide a breeching connection of the same thickness as the stack and weld to[the][each] stack and reinforce the stack externally with steel plates and/or structural members as required to compensate the stack for the structural strength of the metal removed. Provide a flange for bolting breeching to this connection. Provide a hinged cast-iron or steel cleanout door with a heavy duty cast steel frame at the bottom of each stack. Frame and door must be fitted gas-tight. Doors must be a minimum of 457 by 610 mm 18 by 24 inches in size.

2.12.3 Lining

Stack lining must be an alumina-silica-base castable insulating refractory of the hydraulic setting type certified by the manufacturer to be suitable for the specific application, considering the flue gas temperature, velocities, moisture content, and corrosive qualities of the fuel being burned, applied to each stack under the direct supervision of a qualified representative of the manufacturer. The lining thickness must be 76 mm 3 inches. Stack base plate exposed inside of each stack must be covered with castable insulating refractory as specified for stack lining. Boiler stack lining must have a minimum pyrometric cone equivalent (PCE) of 10 1304 degrees C 2380 degrees F with a maximum installed dry weight of 720 kilogram per cubic meter 45 pounds per cubic foot and a recommended service temperature of 1093 degrees C 2000 degrees F. The top 3 meters 10 feet of lining must be a dense castable refractory 1840 kilograms per cubic meter 115 lb/ft 3 suitable for all weather and flue gas temperature conditions encountered in the top of stacks. Stack lining must be fastened to each stack with heat and corrosion resistant alloy metal anchors which maintain their structural strength up to 760 degrees C 1400

degrees F. Anchor clips must be spaced vertically not greater than 610 mm 24 inches on centers and closer horizontally as required by diameter of the stack. Types, size, and number per section must be randomly located by the stack and refractory manufacturer to adequately support refractory lining to permit expansion and prevent cracking of refractory. A 50 mm 2 inch mesh galvanized wire screen of 8 gage (4.10 mm) (0.1620 inch) wire must be attached to the clips for reinforcing the refractory. Locate screen at midpoint of the refractory thickness and cover end of clips with at least 25 mm one inch of refractory.

2.12.4 Stacks

Clean stacks of dirt, rust, oil, and grease by wire brushing and solvent degreasing and given one shop coat of heat-resisting aluminum paint conforming to SAE AMS3126D on the inside and outside. The coat of paint must have a minimum dry film thickness of 0.025 mm one mil.

2.13 BLOWDOWN EQUIPMENT

Furnish[the boiler][the plant] with equipment, tanks, and controls necessary for bottom[and continuous] blowdown of the boiler(s). The equipment for bottom blowdown systems must include a[blowdown tank][and][sample cooler]. Continuous blowdown systems must be of the packaged, proportional type consisting of a heat exchanger, flow control valve, surge tank,[and] sample cooler[, and blowdown control console with test sink]]. Install and pipe blowdown equipment as indicated and it must conform to the recommendations of the NBBI NB-27.

2.13.1 Bottom Blowdown Tank

The blowdown tank must be fabricated of welded steel plate in accordance with ASME BPVC SEC VIII D1. The tank must be a vertical cylindrical tank designed for the working pressure of the boilers. The tank must be equipped with a tangential blowdown inlet located so as to impinge on a 10 mm 3/8 inch carbon steel wear plate extending at least 180 degrees around the interior circumference of the tank from the point of inlet. The tank must be equipped with an internal overflow, vent, drain, safety relief valve, and gage glass with try cocks, blowdown cock, and guard. The interior of the tank must be protected by an epoxy coating system suitable for continuous water immersion and operation at a minimum temperature of [121][149] degrees C [250][300] degrees F. When buried, the tank[must be fitted[with renewable magnesium anodes][with cathodic protection equipment] to minimize galvanic corrosion.][must be constructed of Class A reinforced concrete and must be fitted with a bolted steel manhole frame and cover. Install the blowoff pipe, vent pipe, and drain pipe in pipe sleeves built into the concrete. Fill and caulk the space between the pipe and sleeves with lead wool or equivalent to make a watertight seal. The tank must be divided into two sections by means of a baffle to form a sediment chamber.] Size and locate the blowoff tank as shown.

2.13.2 Sample Cooler

Provide a water-cooled, shell-and-tube or shell-and-coil type heat exchanger designed for cooling sample of boiler water prior to chemical testing. The sample cooler must be furnished as a component of the packaged continuous blowdown system when such a system is being furnished. The cooler must consist of a cast iron or steel shell with copper coil or copper alloy tubes and must be equipped with a brass or bronze sampling cock.[The cooler must be connected to a header and so

valved that a sample can be withdrawn from any boiler as desired.][
Furnish concentrimeter kit containing necessary glassware, reagents, and
instructions for determining boiler water concentrations.]

2.13.3 Continuous Blowdown System

**NOTE: Specify continuous blowdown only where makeup
water ratio is in excess of 20 percent of the boiler
output or where the total dissolved solids of the
makeup water are in excess of 500 ppm.**

**NOTE: Include this requirement if a console type
unit is indicated.**

Provide a complete packaged unit of the automatic-proportioning, [manual-apportioning] type wherein the amount of blowdown from the [boiler] [plant] is automatically proportioned to the amount of make-up feedwater [and the total amount of blowdown from the plant is manually apportioned between boilers according to their steaming rate]. The system must include either an automatic-proportioning valve and a heat exchanger, or a concentric tube automatic proportioning control which must be provided with a separate heat exchanger when necessary to meet the performance requirements indicated or specified herein. The system must be designed for not less than boiler design pressure and must be capable of heating the feedwater from [_____] degrees C F to [_____] degrees C F at the flow rates indicated. The heat exchanger must consist of a steel shell and heads with Type 304 stainless steel tubes arranged in a removable U-bend bundle. The shell must be constructed and tested in accordance with ASME BPVC SEC VIII D1 for the specified boiler operating pressure. The automatic proportioning valve must be provided with a sensing orifice on both the make-up and blowdown lines and must be of the adjustable ratio type in which the ratio of make-up to blowdown may be set anywhere within a range of [30:1][_____] to [4:1][_____]. The automatic proportioning control must consist of two concentric tubes, the inner tube being of a thermostatic design which acts directly against an adjustable seat in response to the temperature differential between the blowdown in the inner tube and the makeup water between the inner and outer tubes. Manual apportioning valves must have bronze bodies with stainless steel seats and disks and must be of the indicating type specifically designed for blowoff service. The blowdown system must be complete with strainers, stop valves, [blowdown meters,] thermometers, and other accessories necessary to form complete packaged units. [The blowdown control console must include illuminated sight flow indicators, automatic flushing and cooling valves, and complete panelboard instrumentation, as well as a cabinet type laboratory sink with drain board, back splash, hot and cold service water faucets, [air cock,] and electrical outlet.]

2.14 FEEDWATER EQUIPMENT

2.14.1 Boiler Feed Pumps

Boiler feed pumps must conform to API STD 610 for motor driven, horizontal split case or support head boiler feed pumps, except as otherwise specified herein. Pumps may be of either the centrifugal or peripheral-turbine type with alloy steel casing and must be bronze or

alloy steel fitted. For centrifugal type pumps, by-pass orifice must be provided, and for turbine type pumps, pressure relief valves must be provided. Packed stuffing boxes or mechanical seals suitable for[121 degrees C 250 degree F operation][the design conditions indicated] must be provided. Pumps must be designed for the net positive suction head, discharge head, and water temperature indicated. Pump casings should be rated at suction head plus shutoff head at design temperature.[Capacity under the above condition must be not less than indicated.][Capacity of each pump under the above conditions must be not less than the following percentage of maximum total boiler capacity: centrifugal pumps, 125 percent; turbine pumps, 150 percent.] Pump motors must be[totally enclosed][dripproof with encapsulated windings].

2.14.2 Boiler Feed Tank

Feed tank and stand construction must conform to ASME BPVC SEC VIII D1 for horizontal cylindrical stand mounted receivers, and must be[hot dip galvanized or cement lined][epoxy coated][[_____] coated]. Tanks must be provided with vents, gage glass, drain and overflow connections, pressure gage, thermometer,[float operated make-up water feeder,][float switch and make-up water solenoid valve,][and preheater assembly consisting of corrosion resistant steam diffuser tube, steam pressure reducing valve, strainer and thermostatic steam valve]. The boiler feed tank assembly must include strainer and pump control box. Tank capacity and connection sizes must be as indicated.

2.14.3 Deaerator

The deaerator must be of the pressurized type having an ASME stamped pressure vessel and conforming to ASME BPVC SEC VIII D1 and requirements specified herein. Deaerating assembly and deaerated water storage may be in the same or separate shells. When external vent condensers are provided, they may be located as recommended by the manufacturer. Provide an ASME certified pressure relief valve sized[as indicated][in accordance with Table II of ASME BPVC SEC VIII D1]. Inlet piping and accessories must be as indicated. Feedwater pumps, as specified herein, interconnecting piping, and control box must be provided as part of the deaerator package. Deaerator capacity must be not less than[indicated][1.25 times that required to supply the boiler(s) at maximum firing rate]. The temperature of the water delivered at maximum capacity must be equivalent to saturated steam temperature at the operating pressure of the deaerator, which must be as indicated, and the oxygen content must not exceed[0.005][0.003] cubic centimeters per liter as determined by the Referee Method A (Colorimetric Indigo Carmine) of ASTM D888 without O2 scavenging chemical addition. Water storage capacity must be sufficient to operate the boilers at maximum capacity for [10][_____] minutes. A power and control panel must be furnished with the package deaerator unit. The panel enclosure must be NEMA ICS 6, Type 12 and must be fabricated and painted in accordance with the manufacturer's standard practice. A separate compartment must be furnished for each feedwater pump and each must include a fused disconnect switch with external operating handle and a magnetic starter with three-leg overload protection. The panel must include signal lights, an audible alarm to signal high and low levels in the deaerator, and required relays, transformers, and manual-automatic switches.

2.14.4 Surge Tank and Transfer System

The condensate storage and surge tank must be a cylindrical welded steel

tank mounted and supported as indicated. Design and construct the tank in accordance with the ASME BPVC for the indicated working pressure. Storage capacity must be[as indicated][sufficient to provide adequate water to the deaerator for [10][_____] minutes of[boiler][plant] operation at maximum capacity]. Inlet connections for condensate and make-up water must be as indicated. Equip the tank with liquid level controllers and valves and alarms as indicated. Equip tank with pressure and temperature gages, water level gage, vent, drain, and overflow. Tank must be[hot dip galvanized or cement lined][epoxy coated][[_____] coated]. The surge tank assembly must include condensate transfer pumps, interconnecting piping including strainer, and control box as indicated. The transfer pumps, except for head and temperature requirements which must be as indicated, must operate continuously with a float controlled flow and must conform to the requirements for boiler feed pumps specified herein.

2.15 COLD WATER CONNECTIONS

Connections must be provided which includes consecutively in line a strainer, reduced pressure principle backflow preventers, and water pressure regulator in that order in the direction of the flow. The reduced pressure principle backflow preventers must be provided as indicated and in compliance with Section 22 00 00 PLUMBING, GENERAL PURPOSE. Cold water fill connections must be made to the water supply system as indicated. Necessary pipe, fittings, and valves required for water connections between the boiler and cold water main must be provided as shown. The pressure regulating valve must be of a type that will not stick or allow pressure to build up on the low side. The valve must be set to maintain a terminal pressure of approximately 35 kPa 5 psi in excess of the static head on the system and must operate within a 15 kPa 2 psi tolerance regardless of cold water supply piping pressure and without objectionable noise under any condition of operation.

2.16 ELECTRIC MOTORS

NOTE: Select standard efficiency for motors used less than 750 hours per year and high efficiency for motors used over 750 hours per year. Packaged boilers should utilize the manufacturer's standard efficiency motor.

Motors which are not an integral part of a packaged boiler must be rated for[standard][high] efficiency service per Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Motors which are an integral part of the packaged boiler system must be the highest efficiency available by the manufacturer of the packaged boiler.

2.17 FITTINGS AND ACCESSORIES

Boiler fittings and accessories must be installed with each boiler in accordance with ASME BPVC SEC IV, unless otherwise specified.

2.17.1 Soot Blowers

NOTE: Soot blowers will normally be required on large water tube units burning No. 5 or 6 fuel oil. Manufacturers of boilers should be consulted to

determine if soot blowers are applicable for the design contemplated. Small units are usually manually cleaned.

Where indicated, each boiler must be provided with soot blowers using[compressed air][steam] as the blowing medium. The soot blower system must be the automatic sequencing and intermittent puff type. The soot blower units must be sequenced automatically using successive steps by their controller, each step involving no more than a 70 kPa 10 psi drop in air pressure at the receiver. After one unit is operated in successive steps through its cycle, the controller must shift the operation to the second soot blower unit, and so on, until all units on that boiler have been operated, after which the controller must be shut down automatically by the sequence controls. The soot blower heads must have elements of suitable material for the highest temperatures encountered in the boiler. The sequence timer must have provision for manual selection of the soot blower units to be used. Soot blower system for oil fired boilers must conform to NFPA 85.

2.17.1.1 Air Compressor Unit

The air compressor unit must conform to ASME PTC 10 except as specified otherwise. Compressor speed must not exceed 900 rpm. Motor speed must not exceed 1750 rpm. The service air requirements must be as indicated with receivers sized as indicated. The units must be suitable for heavy-duty service (soot blowing). The compressors must be simplex type, single-stage, double-acting, with water-jacketed cylinder, fitted with intake and discharge valves of the lightweight feather, disc or plate type, and must be provided with necessary controls, water-cooled aftercooler, moisture separator, drive, receiver, relief valves, and cooling water controls as required. The compressor air intake must be provided with an air suction filter/silencer suitable for outdoor installation. The filter must have a collection efficiency of 99 percent of particles larger than 10 microns. The filter body and media must withstand a pressure of 850 kPa 125 psi. The aftercooler must be the shell-and-tube type designed for air flow through the tubes with steel shell internal baffle plates. The cooling capacity of the after cooler must be sized for the total capacity of the compressor. The moisture separator must be provided with an automatic water discharge trap and level gauge. Cooling water controls for regulating compressor cylinder water temperature and after-cooler water temperature must be thermostatic valve type and must be installed with a three-valve bypass in the water outlet lines ahead of open sight drain funnels. The compressor must be equipped with adjustable, pressure type unloader controls suitable for continuous compressor operation.

2.17.1.2 Air Receiver

The air receiver must be a vertical type constructed in accordance with ASME BPVC SEC VIII D1 for unfired pressure vessels for 1379 kPa 200 psi working pressure, and must be equipped with flanged inlet and outlet connections, valved drain connection, minimum 150 mm 6 inch dial pressure gauge, pop safety valves, and regulator connections.

2.17.2 Continuous Emissions Monitoring

Emerging flue gas flow monitor technologies are

available. The traditional differential pressure technique specified used familiar equipment that can be maintained by plant personnel. This type of measurement device has reliably satisfied regulatory requirements. The possible use of other technologies should include a thorough investigation of flue gas flow monitor regulatory requirements and inhouse maintenance capabilities.

- a. Continuous Emissions Monitoring System (CEMS) equipment must be provided as a system by a single manufacturer. A CEMS, meeting the requirements of applicable federal, State of [____] and local regulations, must be provided for each boiler in accordance with manufacturer's recommendations and under the direct supervision of the CEMS equipment manufacturer. Before acceptance of the installation, the Contracting Officer must be furnished a written test report which provides documentation that the CEMS equipment passed factory and field certification test required by federal, state, and local regulations. Submit written certification by the boiler manufacturer that each boiler furnished complies with Federal, state, and local regulations for emissions. The certification must also include a description of applicable emission regulations. If any boiler is exempt from the emission regulations, the certification must indicate the reason for the exemption.
- b. The reported data must include[sulfur dioxide (SO₂)] [oxides of nitrogen (NO_x)] [carbon dioxide (CO₂)] [and] [particulate matter (PM)] and other information required by Federal, state, and local regulations. SO₂ reporting must be based on[analyzer measurement] [fuel flow and percent sulfur calculation]. Nitrous oxides, carbon dioxide and particulate matter reporting must be based on analyzers.
- c. The CEMS equipment must include the central processing unit, printer, hard disk drive, and floppy disk drive. The floppy disk drive must function as a recorder. The manufacturer must provide the software to generate the required reports in a format acceptable to the Federal, state and local regulatory agencies. The operator interface to the CEMS equipment must be via CRT screen.

2.17.2.1 Gaseous Emission Monitors

Extractive or in situ gaseous monitors must be provided. A combination of extractive and in situ monitors is not acceptable. Gas monitors must include automatic calibration checks. An alarm horn and annunciator must be provided to alarm when any monitor parameter is out of range or a gaseous monitor malfunctions. The surfaces that are exposed to the corrosive gas of the boiler must be constructed of noncorrosive materials such as 316 SS, teflon or hastelloy.

- a. In situ monitor must be mounted on the ductwork at the location[shown on the plans] [recommended by the manufacturer]. The situ system must not be affected by the presence of particulate matter in the flue gas.
- b. Extractive systems must be[wet] [dry] [diluted]. Analyzing equipment for the extractive system must be located in a walk-in cabinet. The equipment must be arranged to provide access for maintenance. Extractive system sampling between the probes and the analyzers must be heat traced to maintain the temperature recommended by the

manufacturer when the ambient temperature is [_____] degrees C F.
Probes must be mounted on the ductwork at the location[shown on the
plans][recommended by the manufacturer].

2.17.2.2 Flue Gas Flow Monitor

Flue gas flow monitor must utilize the pitot tube principle to measure the flow. The probe must be an across-the-duct-average pitot tube and must be designed and located to obtain representative measurement. Differential pressure transmitters must be used to sense the difference between the static and total pressure of the flowing gas steam. Calibrations must be stable. Lines must be arranged to prevent collection of condensate. A purge system must be provided as required to keep the pitot pressure taps clear.

2.17.2.3 Particulate Matter Monitor

Particulate matter (opacity) monitor based on the principle of transmissometry must be provided. The transmissometer must include automatic simulation of zero opacity and upscale check of calibration while the boiler is in service without dismounting the unit. The calibration check must include analyzer internal circuitry and electronic circuitry. An alarm horn and annunciator must be provided to annunciate excess opacity and any system malfunction. Units must be provided with fans to keep the sending and receiving lenses pressurized and blown clean at all times.

2.17.2.4 Wiring

The CEMS equipment must be provided with plug-in prefabricated cable for interconnection between components. Power supply to the equipment must be 2-wire, 120 volt nominal or less, 60 Hz, with one side grounded. Electrical devices must be connected as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

[2.17.3 Tankless Water Heater

**NOTE: If the system will not be used to heat
domestic hot water delete this paragraph.**

A seamless copper immersion type tankless water heater of the specified capacity must be installed in the boiler. The heater must be equipped with an approved water-tempering valve which must be set to supply hot water at approximately 60 degrees C 140 degrees F. Instead of the immersion type coil, an approved external shell and tube type or plate type heat exchanger may be installed as specified in Section 23 57 10.00 10 FORCED HOT WATER HEATING SYSTEMS USING WATER AND STEAM HEAT EXCHANGERS.

]2.17.4 Conventional Breeching and Stacks

2.17.4.1 Breeching

Each boiler must be connected to the stack or flue by breeching constructed of black steel sheets not less than 1.2 mm 0.0478 inch thick nor less than thickness of stack, whichever is larger. Plastic materials polyetherimide (PEI) and polyethersulfone (PES) are forbidden to be used for vent piping of combustion gases. The clear distance between any

portion of the breeching surface and any combustible material must not be less than that specified in NFPA 211. Joints and seams must be securely fastened and made airtight. Suitable hinged and gasketed cleanouts must be provided, which will permit cleaning the entire smoke connection without dismantling. Flexible-type expansion joints must be provided as required and must not require packing.

2.17.4.2 Stacks

NOTE: Frequently boiler outlets are designed to support no more than the weight of a 6 m 20 foot stack section, when installed directly above the boiler outlet. Ensure that the stack is properly supported.

[Individual stub stacks must extend above the roof to the heights indicated. Individual stub stacks must be [6][_____] m [20][_____] feet in height when assembled on the boiler and measured from the ground line. Stack section must be sheet steel having a thickness of not less than 2.47 mm 0.0972 inch. Prefabricated double wall stacks system must extend above the roof to the height indicated. The stacks must be [6][_____] m [20][_____] feet in height when assembled on the boiler and measured from the ground line. The inner stack must be [304 stainless steel] [316 stainless steel] having a thickness of not less than 0.89 mm 0.035 inch. The outer stack must be sheet steel having a thickness of not less than 0.635 mm 0.025 inch. A method of maintaining concentricity between the inner and outer stacks must be incorporated. The joints between the stack sections must be sealed to prevent flue gas leakage.] A 7.92 mm 0.3125 inch diameter hole must be provided in the stack not greater than 150 mm 6 inches from the furnace flue outlet for sampling of the exit gases. A method must be provided to seal the hole to prevent exhaust gases from entering the boiler room when samples are not being taken. Each stack must be provided complete with rain hood. Plastic materials polyetherimide (PEI) and polyethersulfone (PES) are forbidden to be used for vent piping of combustion gases.

2.17.5 Filters

Filters must conform to ASHRAE 52.2.

2.17.6 Foundation (Setting) Materials

2.17.6.1 Firebrick

Firebrick must be ASTM C27 class as recommended by boiler manufacturer.

2.17.6.2 Tile

Tile must be ASTM C34, Grade LBX.

2.17.6.3 Insulating Brick

Insulating brick must comply with ASTM C155.

2.17.6.4 Refractory Mortar

Refractory mortar must comply with ASTM F1097.

2.17.6.5 Castable Refractories

Castable refractories must be [ASTM C401](#). The minimum modulus of rupture for transverse strength must be not less than [4136 kPa 600 psi](#) after being heat soaked for 5 hours or more at a temperature in excess of [1371 degrees C 2500 degrees F](#).

2.17.7 Steel Sheets

2.17.7.1 Galvanized Steel

Galvanized steel must be [ASTM A653/A653M](#).

2.17.7.2 Uncoated Steel

Uncoated steel must be composition, condition, and finish best suited to the intended use.

2.17.8 Gaskets

Gaskets must be nonasbestos material in accordance with [ASME B16.20](#), full face or self-centering type. The gaskets must be of the spiral wound type with graphite filler material.

2.17.9 Dielectric Waterways and Flanges

Dielectric waterways must have temperature and pressure rating equal to or greater than that specified for the connecting piping. Waterways must have metal connections on both ends suited to match connecting piping. Dielectric waterways must include dielectric unions to prevent current flow between dissimilar metals. Dielectric flanges must meet the performance requirements described herein for dielectric waterways.

2.17.10 Flexible Pipe Connectors

Flexible pipe connectors must be designed for [861.8 kPa 125 psi](#) or [1034.2 kPa 150 psi](#) service. Connectors must be installed where indicated. The flexible section must be constructed of rubber, tetrafluoroethylene resin, or corrosion-resisting steel, bronze, monel, or galvanized steel. Materials used and the configuration must be suitable for the pressure, vacuum, and temperature medium. The flexible section must be suitable for service intended and may have threaded, welded, soldered, flanged, or socket ends. Flanged assemblies must be equipped with limit bolts to restrict maximum travel to the manufacturer's standard limits. Unless otherwise indicated, the length of the flexible connectors must be as recommended by the manufacturer for the service intended. Internal sleeves or liners, compatible with circulating medium, must be provided when recommended by the manufacturer. Covers to protect the bellows must be provided where indicated.

2.17.11 Pipe Supports

Pipe supports must conform to [MSS SP-58](#).

2.17.12 Pressure Gauges

Gauges must conform to [ASME B40.100](#) and must be provided with throttling type needle valve or a pulsation dampener and shutoff valve. Minimum dial

size must be 90 mm 3-1/2 inches. A pressure gauge must be provided for each boiler in a visible location on the boiler. Pressure gauges must be provided with readings in kPa psi. Pressure gauges must have an indicating pressure range that is related to the operating pressure of the fluid in accordance with the following table:

Operating Pressure (kPa) (psi)	Pressure Range (kPa) (psi)
519-1030 76-150	0-1400 0-200
105-518 16-75	0-690 0-100
14-104 2-15	0-210 0-30 (retard)

2.17.13 Thermometers

Thermometers must be provided with wells and separable corrosion-resistant steel sockets. Mercury must not be used in thermometers. Thermometers for[inlet water and outlet water for each hot water boiler][the feedwater for each steam boiler] must be provided in a visible location on the boiler. Thermometers must have brass, malleable iron, or aluminum alloy case and frame, clear protective face, permanently stabilized glass tube with indicating-fluid column, white face, black numbers, and a minimum 225 mm 9 inch scale. The operating range of the thermometers must be 0-100 degrees C 32-212 degrees F. The thermometers must be provided with readings in degrees C F.

2.17.14 Air Vents

NOTE: Air vent locations will be indicated on drawings; distinguish between manual and automatic air vents.

2.17.14.1 Manual Air Vents

Manual air vents must be brass or bronze valves or cocks suitable for the pressure rating of the piping system and furnished with threaded plugs or caps.

2.17.14.2 Automatic Air Vents

Automatic air vents must be 19 mm 3/4 inch quick-venting float and vacuum air valves. Each air vent valve must have a large port permitting the expulsion of the air without developing excessive back pressure, a noncollapsible metal float which will close the valve and prevent the loss of water from the system, an air seal that will effectively close and prevent the re-entry of air into the system when subatmospheric pressures prevail therein, and a thermostatic member that will close the port against the passage of steam from the system. The name of the manufacturer must be clearly stamped on the outside of each valve. The air vent valve must be suitable for the pressure rating of the piping system.

2.17.15 Steam Traps

NOTE: The design engineer, when designating steam using equipment or special steam applications, will indicate the type of steam trap required in accordance with the following data:

a. **Inverted Bucket Traps:** This type of trap continuously vents air and carbon dioxide at steam temperature and is recommended for modulating loads. The bucket floats on steam to close the outlet and sinks into condensate to open the condensate outlet. Any trapped air is discharged first into the condensate return line and is followed by condensate discharge. This type of trap has the longest life on systems under modulated control. When large amounts of air are anticipated, an external thermostatic air vent should be installed on a line bypassing the trap to bleed air from the steam line and discharge it to the condensate return line. This system will give optimum performance at low steam pressures with maximum dependability. These traps will handle condensate from fan coil units where condensate must be lifted to return lines located above the equipment. They operate best at near full load conditions where loads do not vary over a wide range. Before operation, traps must be primed by filling them with water.

b. **Vertical Open-Top Bucket Trap:** Trap construction is more complex than inverted bucket type but is suitable for applications having wide variation of load and pressure and is recommended for constant pressure systems. Bucket sinks into condensate when condensate reaches top of trap and the discharge port opens. After discharge, the bucket floats on incoming condensate keeping the discharge port closed.

c. **Impact-Operated Traps:** These traps depend on steam velocity to keep the disc closed. As steam velocity decreases, the disc lifts off the seat and allows flow of condensate. These traps allow some steam leakage and do not vent air at low pressure. They are not recommended for service lower than 69 kPa (10 psig) or where back pressure may exceed 50 percent of inlet pressure. These traps are less expensive and have poor performance in the presence of dirt.

d. **Thermostatic Traps:** These traps are bellows-actuated and contraction of bellows at a few degrees below saturated steam pressure allows condensate air and noncondensable gases to be discharged. As steam reaches the bellows the expansion of the bellows closes the discharge port. These traps can also be utilized to vent air from a steam system and can be used in conjunction with an inverted bucket steam trap previously described.

e. **Float and Thermostatic Trap:** These traps provide optimum performance on modulating systems at lowest first cost. Where steam pressures modulate down to zero, large amounts of air may be liberated. They are ideal for dripping ends of steam risers, heels of up-feed steam risers, bottoms of down-feed stem risers. These traps are also ideal for fan coil units and unit heaters.

f. Any trap selected must be sized for the expected condensate load with an applicable safety factor applied for the particular type of equipment serviced. Manufacturer's application manuals should be consulted to assist in sizing traps. Safety factors vary from 2:1 to 10:1. An average 3:1 safety factor value will cover most applications.

g. Service life between repairs or replacement of traps may be a determining factor in the choice of traps. One manufacturer of all types of traps offers the following experience record:

Type of Trap	Average Service Life Between Replacement or Repairs (months)
Inverted bucket traps	42
Float and thermostatic traps	24
Thermostatic traps	24
Impact-operated traps	19

2.17.15.1 Thermostatic Traps

Thermostatic traps must conform to the requirements of **ASTM F1139** and must be installed in the return connection from each radiator and elsewhere as indicated. Drip traps for mains, risers, and similar lines must be installed with a cooling leg of **1.50 m 5 feet** of uncovered **19 mm 3/4 inch** pipe. The capacity of traps must be based on a pressure differential of **15 kPa 2 psi**. The traps must be designed for a steam working pressure of **100 kPa 15 psig** but must operate with a supply pressure of approximately **15 kPa 2 psig**. The traps must be angle or straight-through pattern with union inlet connections as indicated. The trap bodies and covers must be brass. Valve mechanisms and seats must be monel, stainless steel or hard bronze and must be removable for servicing or replacement.

2.17.15.2 Float-and-Thermostatic Traps

Float-and-thermostatic traps must conform to the requirements of **ASTM F1139** and be designed for a steam working pressure of **100 kPa 15 psig** but must operate with a supply pressure of approximately **34 kPa 5 psig**. The trap capacity must be based on a pressure differential of **15 kPa 2 psig**. Each float-and-thermostatic trap must have a cast iron body and must be provided with a hard bronze, monel, or corrosion-resisting steel valve

seat and mechanism, an open- or closed-type float of brass or equally corrosion-resistant metal, and a corrosion-resisting steel thermostatic air vent, all of which can be easily removed for inspection or replacement without disturbing the piping connections. The inlet to each trap must have a brass or stainless steel strainer, either as an integral part of the trap or as a separate item of equipment.

2.17.15.3 Inverted Bucket Traps

Inverted bucket traps must conform to the requirements of **ASTM F1139** and be designed for a steam working pressure of **100 kPa 15 psig** but must operate with a supply pressure of approximately **35 kPa 5 psig**. Each trap must have a cast iron body and must have a corrosion-resistant steel valve and seat and a brass or corrosion-resistant steel bucket, all of which can be easily removed for inspection or replacement without disturbing the piping connections. The inlet to each trap must have a brass or stainless steel strainer, either as an integral part of the trap or as a separate item of equipment.

2.18 ELECTRICAL EQUIPMENT

NOTE: Select standard efficiency for motors used less than 750 hours per year and high efficiency for motors used over 750 hours per year. The efficiency of each motor will be indicated in the equipment schedules.

Electric motor-driven equipment must be provided complete with motors, motor starters, and necessary control devices. Electrical equipment, motor control devices, motor efficiencies and wiring must be as specified in Section **26 20 00** INTERIOR DISTRIBUTION SYSTEM. Motors which are not an integral part of a packaged boiler and which are integral in size must be the premium efficiency type in accordance with **NEMA MG 1**. Motors which are an integral part of the packaged boiler must be the highest efficiency available by the manufacturer of the packaged boiler. Motor starters must be provided complete with properly sized thermal overload protections and other appurtenances necessary for the motor control specified. Starters must be furnished in[general purpose][watertight][explosion-proof, Class I, division I] enclosures. Manual or automatic control and protective or signal devices required for the operation specified and any control wiring required for controls and devices but not shown must be provided.

2.18.1 Motor Ratings

Motors must be suitable for the voltage and frequency provided. Motors **375 W 1/2 hp** and larger must be three-phase, unless otherwise indicated. Motors must be of sufficient capacity to drive the equipment at the specified capacity without exceeding the nameplate rating on the motor.

2.18.2 Motor Controls

NOTE: The motor controls must be properly coordinated with Section **26 20 00 INTERIOR DISTRIBUTION SYSTEM. Coordinate with the electrical designer for power factors, service factors, and**

desired type of control.

Motor controllers must be provided complete with properly sized thermal overload protection. Manual or automatic control and protective or signal devices required for the operation specified and any wiring required to such devices must be provided. Where two-speed or variable-speed motors are indicated, solid-state variable-speed controllers may be provided to accomplish the same function. Solid state variable speed controllers must be utilized for fractional through 7.46 kW 10 hp ratings. Adjustable frequency drives must be used for larger motors.

2.19 INSULATION

Shop and field-applied insulation must be as specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

2.20 TOOLS

Special tools must be furnished. Special tools must include uncommon tools necessary for the operation and maintenance of boilers, burners, pumps, fans, controls, meters, special piping systems, and other equipment. Small hand tools must be furnished within a suitable cabinet, mounted where directed.

2.20.1 Breeching Cleaner

A cleaner must be provided to clean the breeching. The cleaner must have a jointed handle of sufficient length to clean the breeching without dismantling.

2.20.2 Tube Cleaner

If a watertube boiler is being furnished, a water-driven tube cleaner with three rotary cutters and rotary wire brush complete with the necessary length of armored water hose, valves, and other appurtenances necessary for operation must be provided. Tube cleaner and rotary brush must be provided for each size of water tube in the boiler, with one extra set of cutters for each size cleaner. Necessary valves and fittings must be provided to permit ready connection of the cleaner hose to a high-pressure pump for cold water supply to operate the cleaner.

2.20.3 Tube Brush

If a firetube boiler is being furnished, a tube brush, with steel bristles and jointed handle of sufficient length to clean full length of firetubes, must be provided.

2.20.4 Wrenches

Wrenches must be provided as required for specialty fittings such as manholes, handholes, and cleanouts. One set of extra gaskets must be provided for all manholes and handholes, for pump barrels, and other similar items of equipment. Gaskets must be packaged and properly identified.

2.21 FUEL OIL STORAGE SYSTEM

The fuel oil storage system must be as specified in Section 33 56 10

FACTORY-FABRICATED FUEL STORAGE TANKS unless noted otherwise. A[helical wound coil constructed of 25 mm 1 inch seamless steel tubing][platecoil suction bell heater constructed of carbon steel not lighter than 1.9 mm (14 gauge) 14 gauge] must be provided in each tank for No. 6 fuel oil and installed around the suction end of the oil line. The coil in each tank must have capacity to heat the fuel oil from [] to [] degrees C degrees F, during the maximum demand of all oil burners connected to the tank. The coil must utilize[steam at [] kPa psig][hot water at [] degrees C degrees F] as the heating medium. The heating coil inlet and outlet connections and the fuel-oil suction and return piping connections must be attached to the same tank manway cover. An additional manhole located above the heater must be provided for removal of the heater as a unit.

PART 3 EXECUTION

3.1 INSTALLATION

Arrange work in a neat and orderly manner so that minimum storage space for equipment and material is required at the project site. Install equipment and material in accordance with the best commercial practices. Systems must be neat in appearance, compact, workmanlike in construction and assembly, and installed for continuous service. Parts must be readily accessible for inspection, repair, and renewal. Inspect equipment and material upon delivery and test after installation. Protect material and equipment from the weather. Contractor must repair any damage to equipment, building or worksite that occurs during the execution of his work. Contractor must leave worksite in a condition equal to or better than that existing before the work was started. Maintain manufacturer's recommended clearances around and over equipment, and as required by Part 1 of the NBIC [and][or] local Jurisdictional Code.

3.1.1 Equipment Foundations

Locate as shown and construct of sufficient size and weight, and proper design to preclude shifting of equipment under operating conditions, or under abnormal conditions which could be imposed upon equipment. Foundations must meet requirements of equipment manufacturer. Maintain equipment vibration within acceptable limits, and must be suitably damped and isolated. Grout equipment mounted on concrete foundations before installing piping. Install piping in such a manner as not to place a strain on equipment. When foundations submitted by Contractor shop drawings are different from those shown, submit calculations by the equipment supplier.

3.1.2 Welding

NOTE: Use of the ASME code or the Federal Construction Guide Specification section on welding depends upon the agency's requirements.

NOTE: Insert appropriate Section number in blank below.

Work must be in accordance with[the applicable sections of the

ASME BPVC SEC I][[_____] WELDING] and AWS Z49.1.

3.1.3 Painting

New equipment painting must be factory applied or shop applied, and must be as specified herein, and provided under each individual section.

3.1.3.1 Factory Painting Systems

Manufacturer's standard factory painting systems may be provided subject to certification that the factory painting system applied will withstand 125 hours in a salt-spray fog test, except that equipment located outdoors must withstand[500][3,000] hours in a salt-spray fog test. Salt-spray fog test must be in accordance with ASTM B117, and for that test the acceptance criteria must be as follows: immediately after completion of the test, the paint must show no signs of blistering, wrinkling, or cracking, and no loss of adhesion; and the specimen must show no signs of rust creepage beyond 3 mm 0.125 inch on either side of the scratch mark.

The film thickness of the factory painting system applied on the equipment must not be less than the film thickness used on the test specimen. If manufacturer's standard factory painting system is being proposed for use on surfaces subject to temperatures above 50 degrees C 120 degrees F, the factory painting system must be designed for the temperature service.

3.1.3.2 Shop Painting Systems for Metal Surfaces

Clean, pretreat, prime and paint metal surfaces; except aluminum surfaces need not be painted. Apply coatings to clean dry surfaces. Clean the surfaces to remove dust, dirt, rust, oil and grease by wire brushing and solvent degreasing prior to application of paint, except metal surfaces subject to temperatures in excess of 50 degrees C 120 degrees F must be cleaned to bare metal.

Where more than one coat of paint is specified, apply the second coat after the preceding coat is thoroughly dry. Lightly sand damaged painting and retouch before applying the succeeding coat. Color of finish coat must be aluminum or light gray.

- a. Temperatures Less Than 50 Degrees C 120 Degrees F: Immediately after cleaning, the metal surfaces subject to temperatures less than 50 degrees C 120 degrees F must receive one coat of pretreatment primer applied to a minimum dry film thickness of 0.0076 mm 0.3 mil, one coat of primer applied to a minimum dry film thickness of 0.0255 mm 1 mil; and two coats of enamel applied to a minimum dry film thickness of 0.0255 mm 1 mil per coat.
- b. Temperatures Between 50 and 205 Degrees C 120 and 400 Degrees F: Metal surfaces subject to temperatures between 50 and 205 degrees C 120 and 400 degrees F must receive two coats of 205 degrees C 400 degrees F heat-resisting enamel applied to a total minimum thickness of 0.05 mm 2 mils.
- c. Temperatures Greater Than 205 Degrees C 400 Degrees F: Metal surfaces subject to temperatures greater than 205 degrees C 400 degrees F must receive two coats of 315 degrees C 600 degrees F heat-resisting paint applied to a total minimum dry film thickness of 0.05 mm 2 mils.

Equipment must be factory finished to withstand the intended end use

environment in accordance with the specifications for the particular end item. Field paint equipment not factory finished as specified herein. Retouch damaged areas of factory-finished equipment on which finish has been damaged and then give a complete finish coat to restore the finish to its original condition. The finish coat must be suitable for exposure in the intended end use environment. Spray painting must comply with OSHA 29 CFR 1910-SUBPART D.

3.1.3.3 Cleaning and Application

Remove dirt, rust, oil, and grease by wire brushing and solvent degreasing prior to application of paint. Apply paint to clean and dry surfaces only. Where more than one coat of paint is specified, apply the second coat after the first coat is thoroughly dry. Retouch damaged painting before applying the succeeding coat. Finished surfaces must be smooth. The painting of zinc coated and other corrosion-resistant metal surfaces is not required unless otherwise specified herein.

3.1.3.4 Smoke Flues, Boiler Casing, and Draft Ducts

In unfinished areas, paint smoke flues, boiler casing, and black steel draft ducts with heat-resisting aluminum paint, two coats on the inside of flues and ducts and one coat on the outside, each coat to a minimum dry film thickness of 0.025 mm one mil applied directly to clean bare metal surfaces. Paint exposed surfaces of protective metal covering over insulation, including zinc-coated surfaces, with two coats of heat-resisting black paint to a minimum dry film thickness of 0.05 mm two mils applied directly to the clean bare metal surfaces. Do not paint zinc-coated ducts.

3.1.4 Boiler Cleaning

After installation, boil out[the][each] boiler, under supervision of the manufacturer, with a soda ash or another alkaline cleaning solution to cleanse internal surfaces of oil, grease, mill scale, and dirt. Dispose of the waste products from this process according to the requirements of the applicable federal, state, and local environmental regulations. Following initial cleaning, the boilers should be flushed, drained, washed down (internally and externally) and reinspected to ensure that no traces of oil or foreign matter are present. Next, drain the boiler and associated piping and refill it with treated, softened water. At all times after initial cleaning, the Contractor must protect the boiler, tanks, and piping against internal corrosion until testing is completed and the boiler[s][is][are] accepted. Provide chemicals, labor for introducing chemicals, and professional services for control and supervision of the treatment process.

3.1.5 Marking

Equipment, switches, motor controllers, and other controls or indicating elements not located on the main control panel must be identified by stencils or printed, stamped, or manufactured signs of rigid plastic or non-ferrous material. Lettering for identification signs must be not less than [4.80][] mm [3/16][] inch high. The nomenclature and identification symbols used on the signs must correspond to those used in the maintenance manuals, operating instructions, and schematic diagrams. The signs must be rigidly affixed to the equipment or devices without impairing functions or, if this is not possible, must be attached as a tag using a wire or chain. In addition to the identification signs, each

major component of equipment must have a nameplate listing the manufacturer's name, model number, and when applicable, electrical rating and other information required by pertinent standards or codes.

3.2 EXAMINATION

After becoming familiar with details of the work, verify dimensions in the field, and advise the Contracting Officer of any discrepancy before performing any work or ordering any materials.

3.3 ERECTION OF BOILER AND AUXILIARY EQUIPMENT

NOTE: Consult boiler manufacturers for foundation requirements. Delete the requirement for packing the joint between the boiler and floor with nonasbestos rope, if not required. This packing is typically not required for smaller units.

Boiler and auxiliary equipment must be installed in accordance with manufacturer's written instructions. Proper provision must be made for expansion and contraction between boiler foundation and floor. This joint must be packed with suitable nonasbestos rope and filled with suitable compound that will not become soft at a temperature of **40 degrees C 100 degrees F**. Boilers and firing equipment must be supported from the foundations by structural steel completely independent of all brickwork. Boiler supports must permit free expansion and contraction of each portion of the boiler without placing undue stress on any part of the boiler or setting. Boiler breeching must be as indicated with full provision for expansion and contraction between all interconnected components.

3.4 FIELD QUALITY CONTROL

Performance of tests and inspections as specified herein to demonstrate that boilers and auxiliary equipment, as installed, are in compliance with contract requirements. During **boiler system start-up tests**, factory-trained engineers or technicians employed by manufacturer of such components as the burner, flame safeguard and combustion controls, feedwater treatment equipment, and other auxiliary equipment must be present, as required, to ensure the proper functioning, adjustment, and testing of individual components and systems, and to train plant operation personnel in the operation and maintenance of them. No bypassing, use of jumpers, or other disablement of control systems will be allowed unless specified elsewhere. Labor, equipment, fuel, and test apparatus required for testing must be furnished by the Contractor.

3.4.1 Inspections and Test

NOTE: For Air Force projects, retain the first paragraph in accordance with the latest version of Air Force Manual AFMAN 32-1608. For Army projects, retain the second paragraph in accordance with the latest version of Army Manual AR 420-1. For Navy/Marine projects, retain the third paragraph in accordance with UFC 3-430-07.

- [a. Per latest version of AFMAN 32-1068, the Base Civil Engineer (BCE) must inspect high pressure steam boilers (greater than 105 kPa 15 psig) and HTW generators (greater than 1105 kPa 160 psig and 175 degrees C 350 degrees F) and expansion tanks in active use with NBIC certified inspectors. For low pressure steam boilers (less than 105 kPa 15 psig), hot water boilers (less than 310 kPa 45 psig relief valve), or specially constructed pressure vessels of less than 10 kW 33,500 btu/h in active use, inspection and evaluation must be performed by properly qualified personnel as determined by the BCE or contracted boiler owner/operator.
-] [b. Per latest version of AR 420-1, for high pressure boilers (greater than 105 kPa 15 psig) and all high-temperature water (greater than 102 degrees C 250 degrees F), annual inspections must occur in accordance with NBIC.
-] [c. For all boilers over[117124.2 W 400,000 btu/hr] capacity, per UFC 3-430-07 a NAVFAC Certified Boiler Inspector must perform the boiler inspection of all newly installed boilers.
-] Inspection and testing of boilers and unfired pressure vessels must be made by a boiler inspector certified by one of the NAVFAC Senior Boiler Inspectors. Boilers and direct fired and domestic water heaters under 400,000 BTUs/hour (117124.2 W) input capacity is not covered by UFC 3-430-07 unless requested by the activity.

Inspections and tests at the site must be made under the direction of and be subject to the approval of the Contracting Officer. The Contractor must operate[the][each] boiler and appurtenances prior to final testing and must ensure that necessary adjustments have been made. A[24-][48-] [____]hour advance written notice must be submitted to the Contracting Officer indicating the equipment is ready for testing. Provide testing equipment, including gages, thermometers, calorimeter, CO, O2, CO2 test equipment that independently analyzes each gas, thermocouple pyrometers, fuel flow meters, water meters, and other test apparatus and set up and calibrated prior to the test. Draft, fuel pressure, and steam flow may be measured by permanent gages and meters installed under the contract.[Gas flow may be measured by utility company meters.] Provide an analysis of the fuel being used for tests. Control of noise levels developed by exhaust steam, including muffler, globe, and gate valves, must be conducted in such a manner as not to create a nuisance or hazard, and must be subject to the approval of the Contracting Officer. Tests must include the following, and must be performed when feasible, in the sequence listed:

- a. Strength and tightness tests - Hydro test
- b. Standards compliance tests - All environmental compliance test as required by federal, state and local laws.
- c. Combustion test
- d. Operational tests
- e. Capacity and efficiency tests per ASME PTC 4
- f. Emission test
- g. Tests of auxiliary equipment

h. Feedwater equipment test

3.4.2 Strength and Tightness Tests

Subject [_____] [_____] boilers to the following strength and tightness tests:

3.4.2.1 Hydrostatic Testing

After installation and connection, subject [_____] [_____] boilers to an inspection and hydrostatic test to determine that the boiler and appurtenances have not been damaged in transit or handling. The hydrostatic test must be in accordance with **ASME BPVC SEC I**, with the test pressure applied for a period required by the Contracting Officer. This test must be in addition to the hydrostatic tests performed at the factory. [The hydrostatic test at the site must be certified by an inspector holding an authorized commission from the National Board of Boiler and Pressure Vessel Inspectors.]

3.4.2.2 Pneumatic Testing

Pneumatically test air casing and ducts exterior to the furnace at the maximum working pressure. Use the soap bubble method to verify tightness. Test gas side of boiler normally operated under pressure for tightness at 1-1/2 times the predicted operating pressure in the furnace at the maximum continuous output. For this test, tightly seal the boiler with suitable means to blank off all openings. Admit air to the boiler until the test pressure is reached and then hold. If in a 10-minute period the pressure drop does not exceed **1245 Pa 5 inches water gage**, the casing must be regarded as tight and accepted. Use air pressure and soap bubble tests or comparative carbon dioxide readings for induced draft boilers.

3.4.2.3 Internal Component Test

Hydrostatically test the part of the preassembled fuel oil system that is furnished integrally with the boiler at one and one-half times the maximum operating pressure. Pneumatically test the part of the preassembled gas system that is furnished integrally with the boiler at operating pressure. Use the soap bubble test method to verify tightness of the gas system.

3.4.3 Combustion Tests

Test the fuel burning and combustion control equipment with[the][each of the] specified fuels at the minimum limit of the turndown range and at increments of 50, 75, and 100 percent of full rated load[plus [_____] percent overload]. [The combustion control system must maintain 25 to 35 percent excess air at 20 percent of full rated load, and at 50, 75, and 100 percent of load must maintain an excess air below 15 percent.] Tests must be conducted by factory-trained combustion equipment engineers as previously specified. Analyze and graphically present test data to show for[the][each] boiler at tested loads: rates of steam flow; flue gas temperature; percent excess air; steam quality; and percentages of carbon dioxide, carbon monoxide, and oxygen in the flue gas. Test concentrations of sulfur oxides, particulates, volatiles, and nitrogen oxides in the flue gas to ensure compliance with federal, state, and local environmental requirements. Run tests on each fuel after stack temperatures have stabilized with surroundings. Verify and record conformance to the combustion requirements of this specification. Verify proper operation of instrumentation and gages in the control panel during the test.

3.4.4 Operational Test

Continuously test the boiler(s) continuously under varying load conditions to demonstrate proper operability of the combustion control, flame safeguard control, programming control, and safety interlocks. Conduct this test after the adjustment of the combustion controls has been completed under the combustion test. The operational test must continue for a period of at least [8][____] hours and must include the following:

3.4.4.1 Sequencing

The boiler must start, operate, and stop in strict accordance with the specified operational sequence.

3.4.4.2 Flame Safeguard

Verify the operation of the flame safeguard controls by simulated flame and ignition failures. Test burners having intermittent pilots by simulating main flame failure while the pilot is burning. Verify the trial-for-pilot ignition, trial-for-main flame ignition, combustion control reaction, and valve closing times by stop watch.

3.4.4.3 Immunity to Hot Refractory

Operate the burner at high fire until the combustion chamber refractory reaches maximum temperature. The main fuel valve must then be closed manually. The flame sensing safeguard device must immediately cause the safety shutoff valves to close within the specified control reaction and valve closing times. This test ensures that the flame sensing device will not send a "false positive" signal from glowing refractory.

3.4.4.4 Pilot Intensity Required

Gradually reduce the fuel supply to the pilot flame to the point where the combustion safeguard begins to drop out (sense "no flame") but holds in until the main fuel valve opens. At this point of reduced pilot fuel supply, the pilot flame must be capable of safely igniting the main burner. If the main fuel valve can be opened on a pilot flame of insufficient intensity to safely light the main flame, readjustment of the flame sensing device is required.

3.4.4.5 Boiler Limit and Fuel Safety Interlocks

Safety shutdown must be caused by simulating interlock actuating conditions for high and low temperature, high and low steam pressure, high and low drum level, low feedwater, and fuel safety interlock. Safety shutdowns must occur in the specified manner.

3.4.4.6 Combustion Controls

Demonstrate the accuracy, range, and smoothness of operation of the combustion controls by varying the steam demand through the entire firing range required by the turndown ratio specified for the burner and in case of automatic recycling burners, by further varying the firing rate to require "on-off" cycling. Control accuracy must be as specified.

3.4.4.7 Safety Valves

The high-pressure limit switch must be locked out or otherwise made inoperative, and the boiler safety valves must be lifted by steam. The relieving capacity, popping pressure, blowdown, and reseating pressure must be determined by observation and measurement to be in accordance with the [ASME BPVC SEC I](#). The ASME standard symbol will be accepted only as indicating compliance with the design and material requirements of the code.

3.4.5 Capacity and Efficiency Tests

Perform the capacity and efficiency tests after satisfactory completion of all tests previously specified herein and after the boilers have been operating[continuously] for [1][5][_____] days with no nuisance shutdowns and without the necessity for frequent or difficult adjustments. The contractor is to continuously man the boiler during the test. Perform these tests on each boiler. Conduct tests using[the][each of the] specified fuels. Test procedures must be in accordance with the heat loss method of [ASME PTC 4](#) and must be reported on the ASME Test Form for Abbreviated Efficiency Test. Tests must be performed at[25,] 50, 75 and 100 percent loads. The duration of the tests must be sufficient to record necessary data but in no case must test duration be less than [4][_____] hours[on each fuel].

3.4.6 Auxiliary Equipment and Accessory Tests

Observe and check blowdown valves, stop valves, try cocks, draft fans, oil heaters, pumps, electric motors, and other accessories and appurtenant equipment during the operational and capacity tests for leakage, malfunctions, defects, noncompliance with referenced standards, or overloading, as applicable.

3.4.7 Feedwater Equipment Tests

Perform test of the feedwater treatment equipment in two steps. One test must be conducted by the Contractor concurrently with either the combustion test or the capacity and efficiency test. A second test will be performed by the Government during the first period of heavy loading after the plant has been accepted and put in service. Deficiencies revealed during the Government tests will be corrected under the guarantee provisions of the contract. Both the first and second series of tests must determine compliance with the limits for oxygen content and hardness concentrations of this specification. Equipment for taking samples and the test kit for analyzing the samples must be supplied by the Contractor and must revert to the Government when the tests are completed.

3.4.8 Deaerating Feed-Water Heater

Test of deaerating feed-water heater must demonstrate that the equipment installed must meet the specified requirements as to performance, capacity, and quality of effluent. During the operating test of the boilers, conduct tests to determine oxygen content in accordance with [ASTM D888](#), Method A. Test with no O₂ scavenging chemical addition. Boilers must be operated at varying loads, up to maximum heater capacity, while oxygen tests are being made.

3.4.9 Water Treatment Equipment

Test of water treatment equipment must meet the requirements specified as to capacity and quality of effluent. Tests for ion exchange units must cover at least two complete regenerations and capacity runs. Test for hot process or other precipitation type softeners must cover a minimum continuous period of 48 hours with samples being taken at 2 hour intervals.

3.4.10 Steam Quality

Test for steam quality and water level stability must be simultaneous under the operating conditions specified.

3.4.11 Steam Tests

Tests for boilers over 2068 kPa (gage) 300 psig not used for power generation or large turbine drives and without super-heaters, must be made on steam sampled in accordance with ASTM D1066, and tested for moisture in accordance with the calorimetric method outlined in Section 3, ASME PTC 19.11. Conductivity method may be used in lieu of the calorimetric method, in which case the conductivity of the steam corrected for carbon dioxide and ammonia content must not exceed 30 microhms/cm at 18 degrees C. The steam for boilers less than 2068 kPa (gage) 300 psig used for power generation or turbine drive for air conditioning equipment or with super-heaters, and for boilers over 2068 kPa (gage) 300 psig, must be tested in accordance with the conductivity method in ASTM D2186, with the conductivity of the steam corrected for carbon dioxide and ammonia content not to exceed 4.0 microhms/cm at 18 degrees C.

3.4.12 Water Level Stability Test

Test must first be conducted by use of the manual bypass around the feed-water regulator. Test must be repeated using the automatic feed-water regulator. To be acceptable, the boiler should maintain specified water level stability under both conditions.

3.5 GAS FUEL SYSTEM

Gas piping, fittings, valves, regulators, tests, cleaning, and adjustments must be in accordance with the Section 23 11 20 FACILITY GAS PIPING. Submit proposed test schedules for the heating system and fuel system tests, at least 2 weeks prior to the start of related testing. NFPA 54 must be complied with unless otherwise specified. Burners, pilots, and all accessories must be listed in UL FLAMMABLE & COMBUSTIBLE. The fuel system must be provided with a gas tight, manually operated, UL listed stop valve at the gas-supply connections, a gas strainer, a pressure regulator, pressure gauges, a burner-control valve, a safety shutoff valve suitable for size of burner and sequence of operation, and other components required for safe, efficient, and reliable operation as specified. Approved permanent and ready facilities to permit periodic valve leakage tests on the safety shutoff valve or valves must be provided.

3.6 FUEL OIL SYSTEM

Fuel oil system must be installed in accordance with NFPA 31, unless otherwise indicated.

3.6.1 Piping and Storage Tank

Fuel oil piping and storage tanks must be installed in accordance with Section 33 56 10 FACTORY-FABRICATED FUEL STORAGE TANKS, unless indicated otherwise.

3.6.2 Fuel-Oil Storage Tank Heating-Coil Piping

Supply and return piping and fittings for the heating coil must be installed in accordance with paragraph PIPING INSTALLATION. The[hot water][steam] supply line to the heating coil must be provided with an automatic temperature-control valve, a strainer and a three-valve bypass. The return line from the coil must be provided with a[check valve][steam trap] and a block valve.

3.6.3 Automatic Safety Shutoff Valve

Oil supply line to each oil burner must be equipped with an automatically operated valve designed to shut off the oil supply in case of fire in the immediate vicinity of the burner. The valve must be thermoelectrically actuated or thermomechanically actuated type and must be located immediately downstream of the manual shutoff valve at the day tank inside of the building. If a day tank is not used, the automatic safety valve must be located immediately downstream of the building shutoff devices where oil supply line enters the building. A thermoelectrical or thermomechanical detection device must be located over the oil burner to activate the valve. A fire shutoff valve may be combined with other automatic shutoff devices if listed in UL FLAMMABLE & COMBUSTIBLE.

3.6.4 Earthwork

Excavation and backfilling for tanks and piping must be as specified in Section 31 00 00 EARTHWORK.

3.7 COLOR CODE MARKING AND FIELD PAINTING

NOTE: Designer will coordinate color code marking with Section 09 90 00 PAINTS AND COATINGS. Color code marking for piping not listed in Table 1 of Section 09 90 00 will be added to the table.

Color code marking of piping must be as specified in Section 09 90 00 PAINTS AND COATINGS. Ferrous metal not specified to be coated at the factory must be cleaned, prepared, and painted as specified in Section 09 90 00 PAINTS AND COATINGS. Exposed pipe covering must be painted as specified in Section 09 90 00 PAINTS AND COATINGS. Aluminum sheath over insulation must not be painted.

3.8 MANUFACTURER'S SERVICES

Provide the services of a manufacturer's representative who is experienced in the installation, adjustment, and operation of the equipment specified to supervise the installing, adjusting, and testing of the equipment.

3.9 TEST OF BACKFLOW PREVENTION ASSEMBLIES

Backflow prevention assemblies must be tested in accordance with Section

3.10 HEATING SYSTEM TESTS

NOTE: Whenever possible, the testing of heating systems will be done under adverse winter conditions and low outside temperatures. The test data included will be modified as required to suit the particular heating system.

Select a 4 hour hydrostatic test for radiant floor heating systems in accordance with HYI 400. All other systems should be tested for 2 hours.

Submit the **qualifications** of the firms in charge of installation and testing as specified. Submit a statement from the firms proposed to prepare submittals and perform installation and testing, demonstrating successful completion of similar services of at least five projects of similar size or scope, at least 2 weeks prior to the submittal of any other item required by this section. Before any covering is installed on pipe or heating equipment, the entire heating system's piping, fittings, and terminal heating units must be hydrostatically tested and proved tight at a pressure of 1.5 times the design working pressure, but not less than **689 kPa 100 psi**. Submit proposed test procedures for the heating system tests and fuel system tests, at least 2 weeks prior to the start of related testing.

- a. Before pressurizing system for test, items or equipment (e.g., vessels, pumps, instruments, controls, relief valves) rated for pressures below the test pressure must be blanked off or replaced with spool pieces.
- b. Before balancing and final operating test, test blanks and spool pieces must be removed; and protected instruments and equipment must be reconnected. With equipment items protected, the system must be pressurized to test pressure. Pressure must be held for a period of time sufficient to inspect all welds, joints, and connections for leaks, but not less than 2 hours. No loss of pressure will be allowed. Leaks must be repaired and repaired joints must be retested.
- c. Repair joints must not be allowed under the floor for floor radiant heating systems. If a leak occurs in tubing located under the floor in radiant heating systems, the entire zone that is leaking must be replaced. If any repair is made above the floor for floor radiant heating systems, access must be provided for the installed joint. Caulking of joints must not be permitted.
- d. System must be drained and after instruments and equipment are reconnected, the system must be refilled with service medium and maximum operating pressure applied. The pressure must be held while inspecting these joints and connections for leaks. The leaks must be repaired and the repaired joints retested.

Upon completion of hydrostatic tests and before acceptance of the installation, submit test reports for the heating system tests. Upon completion of testing complete with results, balance the heating system in accordance with Section **23 05 93** TESTING, ADJUSTING, AND BALANCING OF HVAC

SYSTEMS and operating tests required to demonstrate satisfactory functional and operational efficiency. The operating test must cover a period of at least 24 hours for each system, and must include, as a minimum, the following specific information in a report, together with conclusions as to the adequacy of the system:

- a. Certification of balancing.
- b. Time, date, and duration of test.
- c. Outside and inside dry bulb temperatures.
- d. [Temperature of hot water supply leaving boiler][Steam pressure].
- e. Temperature of[heating return water from system at][condensate feed to] boiler inlet.
- f. Quantity of water feed to boiler.
- g. Boiler make, type, serial number, design pressure, and rated capacity.
- h. Fuel burner make, model, and rated capacity; ammeter and voltmeter readings for burner motor.
- i. [Circulating][Condensate][Vacuum]pump make, model, and rated capacity, and ammeter and voltmeter readings for pump motor during operation.
- j. Flue-gas temperature at boiler outlet.
- k. Percent carbon dioxide in flue-gas.
- l. Grade or type and calorific value of fuel.
- m. Draft at boiler flue-gas exit.
- n. Draft or pressure in furnace.
- o. Quantity of water circulated.
- p. Quantity of fuel consumed.
- q. Stack emission pollutants concentration.

Indicating instruments must be read at half-hour intervals unless otherwise directed. Furnish all instruments, equipment, and personnel required for the tests and balancing. Obtain necessary natural gas, water and electricity as specified in the[SPECIAL CONTRACT REQUIREMENTS][Section 01 50 00 TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS]. Provide necessary quantities of propane gas or No. [____] fuel oil when propane gas or fuel oil is require for testing. Operating tests must demonstrate that fuel burners and combustion and safety controls meet the requirements of[ASME CSD-1][ANSI Z21.13/CSA 4.9][NFPA 85].

3.10.1 Water Treatment Testing

The boiler water must be analyzed[prior to the acceptance of the facility][a minimum of once a month for a period of 1 year] by the water treatment company. Submit a water quality test report identifying the

chemical composition of the boiler water. The report must include a comparison of the condition of the boiler water with the manufacturer's recommended conditions. Any required corrective action must be documented within the report. The test report must identify the condition of the boiler at the completion of 1 year of service. The report must include a comparison of the condition of the boiler with the manufacturer's recommended operating conditions. The analysis must include the following information recorded in accordance with **ASTM D596**.

Date of Sample	[_____]
Temperature	[_____] degrees C degrees F
Silica (SiO ₂)	[_____] ppm (mg/1)
Insoluble	[_____] ppm (mg/1)
Iron and Aluminum Oxides	[_____] ppm (mg/1)
Calcium (Ca)	[_____] ppm (mg/1)
Magnesium (Mg)	[_____] ppm (mg/1)
Sodium and Potassium (Na and K)	[_____] ppm (mg/1)
Carbonate (HCO ₃)	[_____] ppm (mg/1)
Sulfate (SO ₄)	[_____] ppm (mg/1)
Chloride (Cl)	[_____] ppm (mg/1)
Nitrate (NO ₃)	[_____] ppm (mg/1)
Turbidity	[_____] ntu
pH	[_____]
Residual Chlorine	[_____] ppm (mg/1)
Total Alkalinity	[_____] epm (meq/1)
Noncarbonate Hardness	[_____] epm (meq/1)
Total Hardness	[_____] epm (meq/1)
Dissolved Solids	[_____] ppm (mg/1)
Fluorine	[_____] ppm (mg/1)
Conductivity	[_____] micro-mho/cm

If the boiler water is not in conformance with the boiler manufacturer's recommendations, the water treatment company must take corrective action.

3.10.2 Boiler/Piping Test

At the conclusion of the 1 year period, the boiler and condensate piping must be inspected for problems due to corrosion and scale. If the boiler is found not to conform to the manufacturer's recommendations, and the water treatment company recommendations have been followed, the water treatment company must provide all chemicals and labor for cleaning or repairing the equipment as required by the manufacturer's recommendations. If corrosion is found within the condensate piping, proper repairs must be made by the water treatment company.

3.11 ACCEPTANCE OPERATIONAL TEST AND INSPECTION

NOTE: For Air Force projects, retain the first paragraph in accordance with the latest version of Air Force Manual AFMAN 32-1608. For Army projects, retain the second paragraph in accordance with the latest version of Army Manual AR 420-1. For Navy/Marine projects, retain the third paragraph in accordance with UFC 3-430-07.

- [a. The Base Civil Engineer (BCE) must inspect high pressure steam boilers (greater than **105 kPa 15 psig**) and HTW generators (greater than **1105 kPa 160 psig** and **175 degrees C 350 degrees F**) and expansion tanks in active use with NBIC certified inspectors. For low pressure steam boilers (less than **105 kPa 15 psig**), hot water boilers (less than **310 kPa 45 psig** relief valve), or specially constructed pressure vessels of less than **10 kW 33,500 btu/h** in active use, inspection and evaluation must be performed by properly qualified personnel as determined by the BCE or contracted boiler owner/operator.
-] [b. Per latest version of AR 420-1, for high pressure boilers (greater than **105 kPa 15 psig**) and all high-temperature water (greater than **102 degrees C 250 degrees F**), annual inspections must occur in accordance with NBIC.
-] [c. For all boilers over[**117124.2 W 400,000 btu/hr**] capacity, per **UFC 3-430-07** a NAVFAC Certified Boiler Inspector must perform the boiler inspection of all newly installed boilers.
-] Prior to requesting an acceptance test, conduct a satisfactory operational test for at least 8 hours, and provide a certified statement that the equipment is installed per all requirements of this guide.

3.12 CLEANING

3.12.1 Boilers and Piping

After the hydrostatic tests have been made and before the system is balanced and operating tests are performed, the boilers and piping must be thoroughly cleaned by filling the system with a solution consisting of either **0.5 kg 1 pound** of caustic soda or **0.5 kg 1 pound** of trisodium phosphate per **190 L 50 gallons** of water. The proper safety precautions must be observed in the handling and use of these chemicals. The water must be heated to approximately **65 degrees C 150 degrees F** and the solution circulated in the system for a period of 48 hours. The system must then be drained and thoroughly flushed out with fresh water.

Strainers and valves must be thoroughly cleaned. Prior to operating tests, air must be removed from all water systems by operating the air vents.

3.12.2 Heating Units

Inside space heating equipment, ducts, plenums, and casing must be thoroughly cleaned of debris and blown free of small particles of rubbish and dust and then vacuum cleaned before installing outlet faces. Equipment must be wiped clean, with all traces of oil, dust, dirt, or paint spots removed. Temporary filters must be provided for fans that are operated during construction, and new provide filters after construction dirt has been removed from the building, and the ducts, plenum, casings, and other items specified have been vacuum cleaned. Perform and document that proper "[Indoor Air Quality During Construction](#)" procedures have been followed; provide documentation showing that after construction ends, and prior to occupancy, new filters were provided and installed. System must be maintained in this clean condition until final acceptance. Bearings must be properly lubricated with oil or grease as recommended by the manufacturer. Belts must be tightened to proper tension. Control valves and other miscellaneous equipment requiring adjustment must be adjusted to setting indicated or directed. Fans must be adjusted to the speed indicated by the manufacturer to meet specified conditions.

3.13 FIELD TRAINING

Conduct a training course for the operating staff as designated by the Contracting Officer. The training period must consist of a total of [____] hours of normal working time and must start after the system is functionally completed but prior to final acceptance tests.

- a. The [field instructions](#) must cover all of the items contained in the approved operation and maintenance manuals, as well as demonstrations of routine maintenance operations and boiler safety devices.
- b. Submit system layout diagrams that show the layout of equipment, piping, and ductwork and typed condensed operation manuals explaining preventative maintenance procedures, methods of checking the system for normal, safe operation, and procedures for safely starting and stopping the system, framed under glass or laminated plastic, at least 2 weeks prior to the start of related testing. After approval, these items must be posted where directed.
- c. Submit [six][____] complete [operation and maintenance instructions](#) listing step-by-step procedures required for system startup, operation, shutdown, and routine maintenance, at least 2 weeks prior to field training. The manuals must include the manufacturer's name, model number, parts list, simplified wiring and control diagrams, troubleshooting guide, and recommended service organization (including address and telephone number) for each item of equipment. Each service organization must be capable of providing [4][____] hour onsite response to a service call on an emergency basis.
- d. Notify the Contracting Officer at least 14 days prior to date of proposed conduction of the training course.

3.14 SCHEDULE

Some metric measurements in this section are based on mathematical

conversion of inch-pound measurements, and not on metric measurements commonly agreed on by the manufacturers or other parties. The inch-pound and metric measurements shown are as follows:

<u>Products</u>	<u>Inch-Pound</u>	<u>Metric</u>
a. [_____]	[_____]	[_____]

-- End of Section --